Exercises 1.5 Functions and their Graphs

1. For $f(x)=x^3+3x$, find each value.

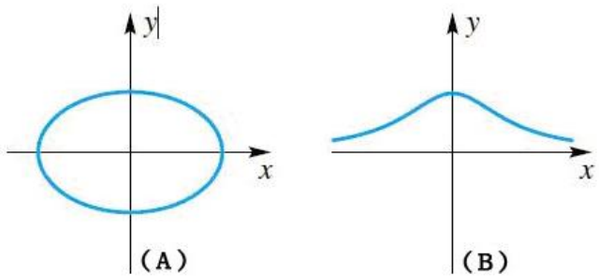
(a) $f(\sqrt{2})$ (b) $f(1+h)$. (c) $f(2)-f(1)$.

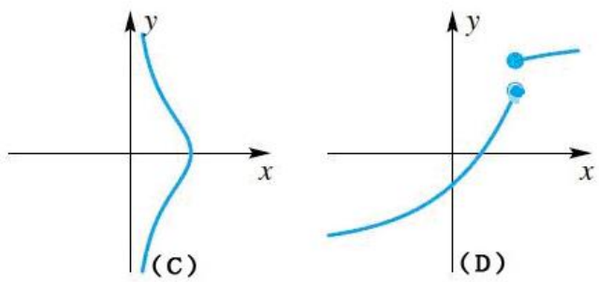
Solution: (a) $f(\sqrt{2})=\left(\sqrt{2}\right)^3+3\sqrt{2}=5\sqrt{2}$

(b) $f(1+h)=(1+h)^3+3(1+h)=h^3+3h^2+6h+4$

(c) $f(2)-f(1)=(2)^3+3(2)-(1)^3-3=14-4=10$.

2. Which of the graphs in the following Figures are graphs of functions? [B]





3. Which of the following functions is **bounded**? [D]
(A function $f(x)$ is called **bounded** if there exists $M>0$ such that $|f(x)|<M$ for all x.)

A. $f(x)=\tan x$ with domain $(0,\frac{\pi}{2})$.

B. $f(x)=\frac{1}{x}$ with domain $(0,\frac{\pi}{2})$.

C. $f(x)=\ln x$ with domain $(0,\frac{\pi}{2})$.

D. $f(x)=x^2$ with domain $(0,\frac{\pi}{2})$.

4. Find the natural domain in each case.

(a) $f(x)=\dfrac{4-x^2}{x^2-x-6}$

Solution: $x^2-x-6=(x+2)(x-3)\neq 0$

So, the natural domain is $\{x\in\mathbb{R}: x\neq -2,3\}$.

(b) $G(y)=\sqrt{(y+1)^{-1}}$

Solution: $(y+1)^{-1}>0\Rightarrow y+1>0\Rightarrow y>-1$

So, the natural domain is $\{y\in\mathbb{R}: y>-1\}$.

(c) $\phi(u)=\ln(2u+3)$

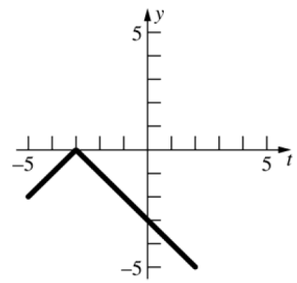
Solution: $2u+3>0\Rightarrow u>-\dfrac{3}{2}$

So, the natural domain is $\left\{u\in\mathbb{R}: u>-\frac{3}{2}\right\}$.

5. Sketch the graphs of the following functions.

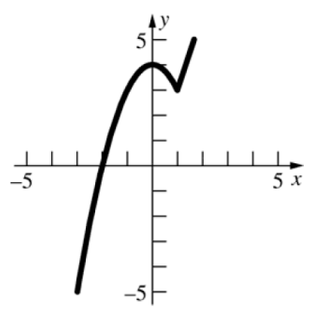
(a) $F(t)=-|t+3|$

Solution:



(b) $h(x)=\begin{cases}-x^2+4 & \text{if } x\le 1\\ 3x & \text{if } x>1\end{cases}$

Solution:

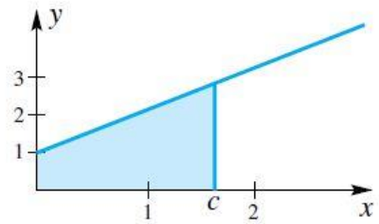


6. Let $A(c)$ denote the area of the region bounded from above by the line $y=x+1$, from the left by the $y-$axis, from below by the $x-$axis, and from the right by the line $x=c$. Such a function is called an *accumulation function* (See Fig.). Find

(a) $A(1)$ (b) $A(2)$ (c) $A(0)$ (d) $A(c)$

(e) Sketch the graph of $A(c)$.

(f) What are the domain and range of A?

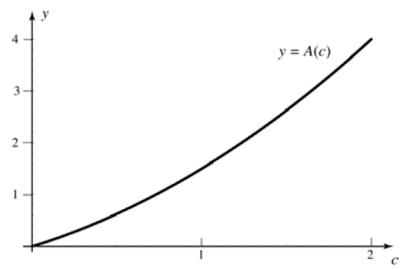


Solution: $A(1)=\frac{1}{2}(1+2)\cdot 1=\frac{3}{2}$;

$A(2)=\frac{1}{2}(1+3)\cdot 2=4$ $A(0)=0$;

$A(c)=\frac{1}{2}(1+1+c)\cdot c=\frac{1}{2}c^2+c$

(e)



(f)Domain: $\{c\in R: c\ge 0\}$ Range: $\{y\in R: y\ge 0\}$

Exercises 1.6 Operations on Functions

1. For $f(x)=x^2+x$ and $g(x)=2/(x+3)$, find each value.

(a) $(f-g)(2)$ (b) $(f/g)(1)$ (c) $g^2(3)$

(d) $(f\circ g)(3)$ (e) $(g\circ f)(1)$ (f) $(g\circ g)(3)$

Solution: (a) $(f-g)(2)=\left(2^2+2\right)-\dfrac{2}{2+3}=\dfrac{28}{5}$

(b) $(f/g)(1)=\dfrac{1^2+1}{\dfrac{2}{1+3}}=4$ (c) $g^2(3)=\left[\dfrac{2}{3+3}\right]^2=\dfrac{1}{9}$

(d) $(f\circ g)(3)=f\left(\dfrac{2}{3+3}\right)=\left(\dfrac{1}{3}\right)^2+\dfrac{1}{3}=\dfrac{4}{9}$

(e) $(g\circ f)(1)=g\left(1^2+1\right)=\dfrac{2}{2+3}=\dfrac{2}{5}$

(f) $(g\circ g)(3)=g\left(\dfrac{2}{3+3}\right)=\dfrac{2}{\dfrac{1}{3}+3}=\dfrac{3}{5}$

2. Find f and g so that $F=g\circ f$.

(a) $F(x)=\sqrt{x+7}$, (b) $F(x)=(x^2+x)^{15}$.

.

Solution: (a) $f(x)=x+7,\ g(x)=\sqrt{x}$

(b) $f(x)=x^2+x,\ g(x)=x^{15}$

3. Let F be any function whose domain contains $-x$ whenever it contains x. Prove each of the following.

(a) $F(x)-F(-x)$ is an odd function.

(b) $F(x)+F(-x)$ is an even function.

(c) F can always be expressed as the sum of an odd and an even function.

Solution: (a) Let $G(x)=F(x)-F(-x)$, then

$G(-x)=F(-x)-F(x)=-G(x)$,hence

$G(x)=F(x)-F(-x)$ is an odd function.

(b) Let $H(x)=F(x)+F(-x)$, then

$H(-x)=F(-x)+F(x)=H(x)$,hence

$G(x)=F(x)-F(-x)$ is an even function.

(c) Since $F(x)-F(-x)$ is an odd function,

$\dfrac{F(x)-F(-x)}{2}$ is also an odd function. Since

$F(x)+F(-x)$ is an even function, $\dfrac{F(x)+F(-x)}{2}$ is also an even function. Therefore,

$F(x)=\dfrac{F(x)-F(-x)}{2}+\dfrac{F(x)+F(-x)}{2}$ the sum of an odd and an even function.

4. Find a formula for $f^{-1}(x)$ of $f(x)=\dfrac{x^3+2}{x^3+1}$ and then verify that $f^{-1}(f(x))=x$ and $f(f^{-1}(x))=x$.

Solution: Let $y=\dfrac{x^3+2}{x^3+1}$. Then $yx^3+y=x^3+2$. So,

$x=\sqrt[3]{\dfrac{2-y}{y-1}}$,that is, $f^{-1}(x)=\sqrt[3]{\dfrac{2-x}{x-1}}$.

$$f^{-1}(f(x))=f^{-1}\left(\frac{x^3+2}{x^3+1}\right)=\sqrt[3]{\frac{2-\left(\frac{x^3+2}{x^3+1}\right)}{\left(\frac{x^3+2}{x^3+1}\right)-1}}=\sqrt[3]{x^3}=x$$

$$f(f^{-1}(x))=f(\sqrt[3]{\frac{2-x}{x-1}})=\frac{(\sqrt[3]{\frac{2-x}{x-1}})^3+2}{(\sqrt[3]{\frac{2-x}{x-1}})^3+1}=x.$$

Exercises 1.9 The Inverse Trigonometric Functions

1. Find the value without using a calculator.

(a) $\arccos(\dfrac{\sqrt{2}}{2})$ (b) $\arcsin(-\dfrac{\sqrt{3}}{2})$

(c) $\arctan(\sqrt{3})$.

Solution: (a) $\arccos(\dfrac{\sqrt{2}}{2})=\dfrac{\pi}{4}$, since $\cos\dfrac{\pi}{4}=\dfrac{\sqrt{2}}{2}$

(b) $\arcsin(-\dfrac{\sqrt{3}}{2})=-\dfrac{\pi}{3}$

(c) $\arctan(\sqrt{3})=\dfrac{\pi}{3}$

2. Show that each equation is an identity.

(a) $\sin(\tan^{-1}x)=\dfrac{x}{\sqrt{1+x^2}}$

Proof: Let $\theta=\arctan x$, then $\tan\theta=x$, $-\pi/2<\theta<\pi/2$.

Then $\cos\theta=\dfrac{1}{\sec\theta}=\dfrac{1}{\sqrt{1+\tan^2\theta}}=\dfrac{1}{\sqrt{1+x^2}}$.

Method 1: When $x\ge 0$, then $0\le\theta<\pi/2$, thus

$\sin(\tan^{-1}x)=\sin\theta=\sqrt{1-\cos^2\theta}=\dfrac{x}{\sqrt{1+x^2}}$.

When $x<0$, then $-\pi/2<\theta<0$, thus

$\sin(\tan^{-1}x)=\sin\theta=-\sqrt{1-\cos^2\theta}=-\dfrac{|x|}{\sqrt{1+x^2}}=\dfrac{x}{\sqrt{1+x^2}}$.

Method 2: $\sin(\tan^{-1}x)=\sin\theta=\tan\theta\cdot\cos\theta=\dfrac{x}{\sqrt{1+x^2}}$.

(b) $\cos(2\sin^{-1}x)=1-2x^2$

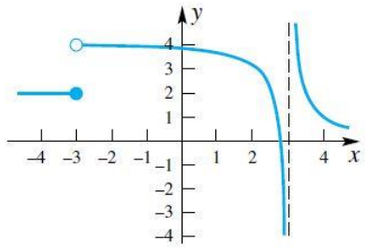
Proof: $\cos(2\sin^{-1}x)=1-2\sin^2\left(\sin^{-1}x\right)=1-2x^2$.

Exercises 2.1 Introductions to Limits

1. For the function f graphed in the Figure , find the indicated limit or function value, or state that it does not exist.

(a) $f(-3)$ (b) $f(3)$ (c) $\lim_{x\to -3^-} f(x)$

(d) $\lim_{x\to -3^+} f(x)$ (e) $\lim_{x\to -3} f(x)$ (f) $\lim_{x\to 3^+} f(x)$



Solution:

(a) $f(-3)$ =2

(b) $f(3)$ does not exist.

(c) $\lim_{x\to -3^-} f(x)$ =2

(d) $\lim_{x\to -3^+} f(x)$ =4

(e) $\lim_{x\to -3} f(x)$ does not exist.

(f) $\lim_{x\to 3^+} f(x)$ does not exist.

Exercises 2.2 Rigorous Study of Limits

1. Let $f(x)=x^2$.

(a) Find a positive number δ_1 such that if $|x-2|<\delta_1$, then $|f(x)-4|<0.1$. Prove your statement.

(b) Find another positive number δ_2 such that if $|x-2|<\delta_2$, then $|f(x)-4|<0.01$. Prove your statement.

Solution: $|x^2-4|=|x-2||x+2|$. If $|x-2|<1$, then $|x+2|<5$. So, $|x^2-4|<5|x-2|$.

(1) Let $\delta_1=0.02$. If $|x-2|<\delta_1$ <1, then $|f(x)-4|<5|x-2|<5\times 0.02=0.1$.

(2) Let $\delta_2=0.002$. If $|x-2|<\delta_2$ <1, then $|f(x)-4|<5|x-2|<5\times 0.002=0.01$.

2. Find a counterexample to prove the following statement is not correct.

"if $f(x)>0$ for all x in some deleted interval about a and $\lim_{x\to a} f(x)$ exists, then $\lim_{x\to a} f(x)>0$."

Solution: Let $f(x)=x^2$. In the deleted interval $(-1,1)\setminus\{0\}$ about 0, $f(x)>0$ and $\lim_{x\to 0} f(x)=0$ exists. But 0 is not greater than 0.

3. Prove that $\lim_{x\to 1} 3x=3$ by the definition.

Proof: For any $\varepsilon>0$, there exists $\delta=\frac{\varepsilon}{3}>0$, such that if $0<|x-1|<\delta$, then $|3x-3|=3|x-1|<3\delta=\varepsilon$.

4. Prove that the limit of $f(x)=\frac{|x|}{x}$ as x approaches 0 does not exist.

Proof: The natural domain of $f(x)=\frac{|x|}{x}$ is $\{x\in\mathbb{R}: x\neq 0\}$. Since $f(x)=1$ for $x>0$ and $f(x)=-1$ for $x<0$, $\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+} 1=1$ and $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^-} -1=-1$.

Since $\lim_{x\to 0^+} f(x)\neq \lim_{x\to 0^-} f(x)$, the limit of $f(x)=\frac{|x|}{x}$ as x approaches 0 does not exist.

Exercises 2.3 Limit Theorems

1. Find the limits if $\lim_{x\to a} f(x)=3$ and $\lim_{x\to a} g(x)=-1$.

(a) $\lim_{x\to a}\frac{2f(x)-3g(x)}{f(x)+g(x)}$ (b) $\lim_{x\to a}\sqrt[3]{g(x)}[f(x)+3]$.

Solution:

(a) $\lim_{x\to a}\frac{2f(x)-3g(x)}{f(x)+g(x)}=\frac{\lim_{x\to a}[2f(x)-3g(x)]}{\lim_{x\to a}[f(x)+g(x)]}$

$=\frac{2\lim_{x\to a}f(x)-3\lim_{x\to a}g(x)}{\lim_{x\to a}f(x)+\lim_{x\to a}g(x)}=\frac{2\cdot 3-3\cdot(-1)}{3-1}=\frac{9}{2}$

(b) $\lim_{x\to a}\sqrt[3]{g(x)}[f(x)+3]$.

$=\sqrt[3]{\lim_{x\to a}g(x)}\cdot[\lim_{x\to a}f(x)+3]=\sqrt[3]{-1}\cdot(3+3)=-6$.

2. Find $\lim_{x\to 2}\frac{f(x)-f(2)}{x-2}$ for $f(x)=3x^2$.

Solution:

$\lim_{x\to 2}\frac{f(x)-f(2)}{x-2}=\lim_{x\to 2}\frac{3x^2-3(2)^2}{x-2}=\lim_{x\to 2}\frac{3(x^2-4)}{x-2}$

$=\lim_{x\to 2}\frac{3(x-2)(x+2)}{x-2}=3\lim_{x\to 2}(x+2)=3(2+2)=12$.

3. Find $\lim_{x\to 1}\frac{x^3-1}{\sqrt{2x+2}-2}$.

Solution:

$\lim_{x\to 1}\frac{x^3-1}{\sqrt{2x+2}-2}=\lim_{x\to 1}\frac{(x^3-1)(\sqrt{2x+2}+2)}{2(x-1)}$

$=\lim_{x\to 1}\frac{(x^2+x+1)(\sqrt{2x+2}+2)}{2}$

$=3\cdot 2=6.$

4. If $\lim_{x\to 1}\frac{x^2+ax+b}{x-1}=2$. Find the values of a and b.

Solution: Since $\lim_{x\to 1}x-1=0$,

$\lim_{x\to 1}x^2+ax+b=0\Rightarrow 1+a+b=0\Rightarrow b=-1-a$

$\lim_{x\to 1}\frac{x^2+ax+b}{x-1}=\lim_{x\to 1}\frac{x^2+ax-a-1}{x-1}=\lim_{x\to 1}\frac{x^2-1+a(x-1)}{x-1}$

$=\lim_{x\to 1}(x+1)+a=2+a=2$.

Therefore, $a=0$ then $b=-1$.

5. Suppose that $f(x)g(x)=1$ for all x and $\lim_{x\to a}g(x)=0$. Prove that $\lim_{x\to a}f(x)$ does not exist.

Proof: Since $f(x)g(x)=1$ for all x, then $\lim_{x\to a}f(x)g(x)=1$. Assume that $\lim_{x\to a}f(x)$ exists. Then $\lim_{x\to a}f(x)g(x)=\lim_{x\to a}f(x)\cdot\lim_{x\to a}g(x)=0$. It's a contradiction.

Thus $\lim_{x\to a}f(x)$ does not exist.

6. . Find each of the following limits or state that it does not exist.

(a) $\lim_{t\to -1}(t^2-x^2)$

Solution: $\lim_{t\to -1}(t^2-x^2)=1-x^2$.

(b) $\lim_{h\to 0}\frac{(x+h)^2-x^2}{h}$

Solution:

$\lim_{h\to 0}\frac{(x+h)^2-x^2}{h}=\lim_{h\to 0}\frac{x^2+2xh+h^2-x^2}{h}=\lim_{h\to 0}\frac{2xh+h^2}{h}$

$=\lim_{h\to 0}(2x+h)=2x+0=2x.$

(c) $\lim_{x\to 3^+}\frac{x-3}{\sqrt{x^2-9}}$

Solution: $\lim_{x\to 3^+}\frac{x-3}{\sqrt{x^2-9}}$

$=\lim_{x\to 3^+}\frac{x-3}{\sqrt{(x+3)(x-3)}}=\lim_{x\to 3^+}\frac{\sqrt{x-3}}{\sqrt{x+3}}=\frac{0}{\sqrt{6}}=0$.

(d) $\lim_{x\to -1^-}\frac{\sqrt{x^2+2x+1}}{x^2-1}$

Solution:

$\lim_{x\to -1^-}\frac{\sqrt{x^2+2x+1}}{x^2-1}=\lim_{x\to -1^-}\frac{\sqrt{(x+1)^2}}{x^2-1}=\lim_{x\to -1^-}\frac{|x+1|}{x^2-1}$

$=\lim_{x\to -1^-}\frac{-(x+1)}{(x+1)(x-1)}=\lim_{x\to -1^-}(-\frac{1}{x-1})=\frac{1}{2}$

(e) $\lim_{x\to 1}\frac{|x-1|}{x-1}$

Solution: $\lim_{x\to 1^-}\frac{|x-1|}{x-1}=\lim_{x\to 1^-}\frac{-(x-1)}{x-1}=-1$ and

$\lim_{x\to 1^+}\frac{|x-1|}{x-1}=\lim_{x\to 1^+}\frac{x-1}{x-1}=1$. Since $\lim_{x\to 1^-}\frac{|x-1|}{x-1}\neq\lim_{x\to 1^+}\frac{|x-1|}{x-1}$,

$\lim_{x\to 1}\frac{|x-1|}{x-1}$ does not exist.

Exercises 2.4 Limits at infinity; Infinite limits

In Problem 1-5, find the indicated limit or state that it does not exist. Justify your statement.

1 $\lim\limits_{x\to\infty}\dfrac{x}{x-5}$.

Solution: $\lim\limits_{x\to\infty}\dfrac{x}{x-5}=\lim\limits_{x\to\infty}\dfrac{1}{1-\dfrac{5}{x}}=1$.

2 $\lim\limits_{x\to 0}\dfrac{1}{x}$

Solution: $\lim\limits_{x\to 0^+}\dfrac{1}{x}=\infty$ and $\lim\limits_{x\to 0^-}\dfrac{1}{x}=-\infty$ .Since

$\lim\limits_{x\to 0^-}\dfrac{1}{x}\neq\lim\limits_{x\to 0^+}\dfrac{1}{x}$, $\lim\limits_{x\to 0}\dfrac{1}{x}$ does not exist.

3 $\lim\limits_{x\to\infty}\sqrt{\dfrac{x^2+x+3}{(x-1)(x+1)}}$.

Solution: $\lim\limits_{x\to\infty}\sqrt{\dfrac{x^2+x+3}{(x-1)(x+1)}}$.

$$=\sqrt{\lim_{x\to\infty}\frac{x^2+x+3}{x^2-1}}=\sqrt{\lim_{x\to\infty}\frac{1+\frac{1}{x}+\frac{3}{x^2}}{1-\frac{1}{x^2}}}=\sqrt{1}=1$$

4 $\lim\limits_{x\to-\infty}\dfrac{2x+1}{\sqrt{x^2+3}}$

Solution:

$$\lim_{x\to-\infty}\frac{2x+1}{\sqrt{x^2+3}}=\lim_{x\to-\infty}\frac{-\frac{1}{x}(2x+1)}{-\frac{1}{x}\sqrt{x^2+3}}=\lim_{x\to-\infty}\frac{-2-\frac{1}{x}}{\sqrt{1+\frac{3}{x^2}}}=-2.$$

5 $\lim\limits_{x\to\infty}(\sqrt{2x^2+3}-\sqrt{2x^2-5})$.

Solution:

$$\lim_{x\to\infty}(\sqrt{2x^2+3}-\sqrt{2x^2-5})$$

$$=\lim_{x\to\infty}\frac{\left(\sqrt{2x^2+3}-\sqrt{2x^2-5}\right)\left(\sqrt{2x^2+3}+\sqrt{2x^2-5}\right)}{\left(\sqrt{2x^2+3}+\sqrt{2x^2-5}\right)}$$

$$=\lim_{x\to\infty}\frac{2x^2+3-\left(2x^2-5\right)}{\left(\sqrt{2x^2+3}+\sqrt{2x^2-5}\right)}=\lim_{x\to\infty}\frac{8}{\left(\sqrt{2x^2+3}+\sqrt{2x^2-5}\right)}$$

$$=\lim_{x\to\infty}\frac{\frac{x}{\sqrt{2x^2+3}+\sqrt{2x^2-5}}}{\sqrt{x^2}}=\lim_{x\to\infty}\frac{\frac{x}{\sqrt{2+\frac{3}{x^2}}+\sqrt{2-\frac{5}{x^2}}}}{}=0$$

Exercises 2.5 Limits involving trigonometric functions

1. Evaluate each limit.

(a) $\lim\limits_{\theta\to 0}\dfrac{\sin 3\theta}{2\theta}$

Solution:

$$\lim_{\theta\to 0}\frac{\sin 3\theta}{2\theta}=\lim_{\theta\to 0}\frac{3}{2}\cdot\frac{\sin 3\theta}{3\theta}=\frac{3}{2}\lim_{\theta\to 0}\frac{\sin 3\theta}{3\theta}=\frac{3}{2}\cdot 1=\frac{3}{2}.$$

(b) $\lim\limits_{\theta\to 0}\dfrac{\tan 5\theta}{\sin 2\theta}$

Solution:

$$\lim_{\theta\to 0}\frac{\tan 5\theta}{\sin 2\theta}=\lim_{\theta\to 0}\left[\frac{\tan 5\theta}{5\theta}\cdot 5\cdot\frac{1}{2}\cdot\frac{2\theta}{\sin 2\theta}\right]$$

$$=\frac{5}{2}\lim_{\theta\to 0}\frac{\tan 5\theta}{5\theta}\lim_{\theta\to 0}\frac{2\theta}{\sin 2\theta}=\frac{5}{2}\cdot 1\cdot 1=\frac{5}{2}.$$

(c) $\lim\limits_{\theta\to 0}\dfrac{\cot(\pi\theta)\sin\theta}{2\sec\theta}$

Solution:

$$\lim_{\theta\to 0}\frac{\cot(\pi\theta)\sin\theta}{2\sec\theta}=\lim_{\theta\to 0}\frac{\frac{\cos(\pi\theta)}{\sin(\pi\theta)}\sin\theta}{\frac{2}{\cos\theta}}=\lim_{\theta\to 0}\frac{\cos(\pi\theta)\sin\theta\cos\theta}{2\sin(\pi\theta)}$$

$$=\lim_{\theta\to 0}\left[\frac{\cos(\pi\theta)\cos\theta}{2}\cdot\frac{\sin\theta}{\theta}\cdot\frac{1}{\pi}\cdot\frac{\pi\theta}{\sin(\pi\theta)}\right]$$

$$=\frac{1}{2\pi}\lim_{\theta\to 0}\left[\cos(\pi\theta)\cos\theta\cdot\frac{\sin\theta}{\theta}\cdot\frac{\pi\theta}{\sin(\pi\theta)}\right]$$

$$=\frac{1}{2\pi}\cdot 1\cdot 1\cdot 1\cdot 1=\frac{1}{2\pi}.$$

2. Use the squeeze theorem to determine $\lim\limits_{x\to 0}x\sin\dfrac{1}{x}$.

Solution: Since $-1\le\sin(1/x)\le 1$ for all x, we have $-|x|\le x\sin(1/x)\le|x|$. And $\lim\limits_{x\to 0}\left(-|x|\right)=\lim\limits_{x\to 0}|x|=0$.

Then by Squeeze Theorem, $\lim\limits_{x\to 0}x\sin\left(\dfrac{1}{x}\right)=0$.

3. Find the limits.

(a) $\lim\limits_{x\to a}\dfrac{\sin x-\sin a}{\sin(x-a)}$ (b) $\lim\limits_{x\to 2}\dfrac{\sin(x-2)}{x^2-4}$

Solution: $\lim\limits_{x\to a}\dfrac{\sin x-\sin a}{\sin(x-a)}$

$$=\lim_{x\to a}\frac{2\cos\frac{x+a}{2}\sin\frac{x-a}{2}}{2\sin\frac{x-a}{2}\cos\frac{x-a}{2}}=\lim_{x\to a}\frac{\cos\frac{x+a}{2}}{\cos\frac{x-a}{2}}=\cos a.$$

(b) $\lim\limits_{x\to 2}\dfrac{\sin(x-2)}{x^2-4}$

Solution: $\lim\limits_{x\to 2}\dfrac{\sin(x-2)}{x^2-4}=\lim\limits_{x\to 2}\dfrac{\sin\left(x-2\right)}{\left(x+2\right)\left(x-2\right)}$

$$=\lim_{x\to 2}\frac{\frac{\sin\left(x-2\right)}{x-2}}{x+2}=\frac{\lim\limits_{x\to 2}\frac{\sin\left(x-2\right)}{x-2}}{\lim\limits_{x\to 2}x+2}=\frac{1}{4}.$$

Exercises 2.6 Natural exponential, Natural log and hyperbolic functions

1. Use the fact that $e = \lim_{h\to 0}(1+h)^{1/h}$ to find each limit.

(a) $\lim_{x\to 0}(1-x)^{1/x}$

Solution:

$$\lim_{x\to 0}(1-x)^{1/x} = \lim_{x\to 0}\frac{1}{\left[1+(-x)\right]^{\frac{1}{-x}}} = \frac{1}{e}.$$

(b) $\lim_{x\to 0}(1+3x)^{1/x}$

Solution:

$$\lim_{x\to 0}(1+3x)^{1/x} = \lim_{x\to 0}\left[(1+3x)^{\frac{1}{3x}}\right]^3 = e^3.$$

(c) $\lim_{x\to\infty}\left(\frac{x+2}{x}\right)^x$

Solution:

$$\lim_{x\to\infty}\left(\frac{x+2}{x}\right)^x = \lim_{x\to\infty}\left(1+\frac{2}{x}\right)^x = \lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{\frac{x}{2}\cdot 2} = e^2.$$

(d) $\lim_{x\to\infty}\left(\frac{x-1}{x}\right)^2$

Solution:

$$\lim_{x\to\infty}\left(\frac{x-1}{x}\right)^2 = \lim_{x\to\infty}\left(1-\frac{1}{x}\right)^2 = 1^2 = 1$$

(e) $\lim_{x\to\infty}\left(\frac{1+x}{x}\right)^{3x}$

Solution:

$$\lim_{x\to\infty}\left(\frac{1+x}{x}\right)^{3x} = \lim_{x\to\infty}\left[\left(1+\frac{1}{x}\right)^x\right]^3 = e^3.$$

(f) $\lim_{x\to\infty}\left(\frac{2x+3}{2x+1}\right)^{3x+1}$.

Solution:

$$\lim_{x\to\infty}\left(\frac{2x+3}{2x+1}\right)^{3x+1} = \lim_{x\to\infty}\left(1+\frac{2}{2x+1}\right)^{3x+1} = \lim_{x\to\infty}\left(1+\frac{1}{x+\frac{1}{2}}\right)^{3\left(x+\frac{1}{2}\right)-\frac{1}{2}}$$

$$= \lim_{x\to\infty}\left(1+\frac{1}{x+\frac{1}{2}}\right)^{3\left(x+\frac{1}{2}\right)} \lim_{x\to\infty}\left(1+\frac{1}{x+\frac{1}{2}}\right)^{-\frac{1}{2}} = e^3\times 1 = e^3.$$

(g) $\lim_{x\to 1^+} x^{\frac{1}{x-1}}$

Solution: (Let $x = 1 + t$.)

$$\lim_{x\to 1^+} x^{\frac{1}{x-1}} = \lim_{t\to 0^+}(1+t)^{\frac{1}{t}} = e$$

(h) $\lim_{x\to\infty}\left(\frac{x^2}{(x-1)(x+3)}\right)^x$.

Solution:

$$\lim_{x\to\infty}\left(\frac{x^2}{(x-1)(x+3)}\right)^x = \lim_{x\to\infty}\left(\frac{x}{x-1}\right)^x\cdot\left(\frac{x}{x+3}\right)^x$$

$$= \lim_{x\to\infty}\left(1+\frac{1}{x-1}\right)\cdot\lim_{x\to\infty}\left(1+\frac{1}{x-1}\right)^{x-1}\cdot$$

$$\lim_{x\to\infty}\left(\left(1+\frac{-3}{x+3}\right)^{\frac{x+3}{-3}}\right)^{-3}\cdot\lim_{x\to\infty}\left(1+\frac{-3}{x+3}\right)^{-3}$$

$$= 1\cdot e^1\cdot e^{-3}\cdot 1 = e^{-2}.$$

2. If \$375 is put in the bank today, what will it be worth at the end of 2 years if interest is 3.5% and is compounded as specified?

(a) Annually (b) Monthly (c) Continuously.

Solution:

(a) $(\$375)(1.035)^2 \approx \$401.71$

(b) $(\$375)(1+\frac{0.035}{12})^{24} \approx \$402.15$

(c) $(\$375)e^{0.035\cdot 2} \approx \$402.19$

Exercises 2.7 Continuity of functions

1. (1) Find the natural domain of $\phi(x)=\dfrac{x^4+2x^2-3}{x+1}$.

(2) Find a function $f(\mathrm{x})$ such that $f(\mathrm{x})$ is continuous in $\mathbb{R}$ and $f(\mathrm{x})=\phi(\mathrm{x})$ in the natural domain of $\phi(\mathrm{x})$.

Solution: (1) Since the denominator $x+1$ cannot be 0, the natural domain of $\phi(x)$ is $\{x\in\mathbb{R}: x\neq -1\}$.

(2)

$$\lim_{x\to-1}\frac{x^4+2x^2-3}{x+1}=\lim_{x\to-1}\frac{(x+1)(x-1)(x^2+3)}{x+1}$$
$$=\lim_{x\to-1}\left[(x-1)(x^2+3)\right]=-8.$$

Let $f(x)=\begin{cases}-8 & \text{for } x=-1\\ \phi(x) & \text{otherwise.}\end{cases}$ Then $f(x)=\phi(x)$ is continuous for $x\neq-1$. Since

$\lim_{x\to-1}f(x)=\lim_{x\to-1}\dfrac{x^4+2x^2-3}{x+1}=-8=f(-1)$, $f(x)$ is also continuous at $x=-1$. So, $f(\mathrm{x})$ is continuous in $\mathbb{R}$.

2. At what points, if any, are the functions discontinuous?

(a) $f(x)=\dfrac{3x+7}{(x-30)(x-\pi)}$

(b) $f(x)=\begin{cases}x & \text{if } x<0\\ x^2 & \text{if } 0\le x\le 1\\ 2-x & \text{if } x>1\end{cases}$

Solution: (a) Function $f(x)$ is discontinuous at $x=\pi, 30$ since $f(x)$ is not well-defined there.

(b) Function $f(x)$ is continuous on $(-\infty,0)$, $(0,1)$ and $(1,\infty)$ as polynomials.

At point $x=0$, $\lim_{x\to0^+}f(x)=\lim_{x\to0^-}f(x)=f(0)=0$ and so $f(x)$ is continuous at $x=0$.

At point $x=1$, $\lim_{x\to1^+}f(x)=\lim_{x\to1^-}f(x)=f(1)=1$ and so $f(x)$ is continuous at $x=1$.

To conclude, $f(x)$ is continuous on $(-\infty,\infty)$.

3. Determine whether the function is continuous at the origin. If the function is not continuous, determine whether the discontinuity is removable or non-removable.

(a) $g(x)=\begin{cases}\dfrac{\sin x}{x} & x\neq0\\ 0 & x=0\end{cases}$,

Solution: Since $\lim_{x\to0}g(x)=\lim_{x\to0}\dfrac{\sin x}{x}=1\neq0=g(0)$, $g(x)$ is discontinuous at $x=0$.

Since $\lim_{x\to0}g(x)=1$ *exists*, $x=0$ is a removable discontinuity.

(b) $F(x)=x\sin\dfrac{1}{x}$.

Solution: Function F(x) is discontinuous at x=0 since $x=0$ is not in the domain of F(x).

Since $\lim_{x\to0}F(x)=\lim_{x\to0}x\sin\dfrac{1}{x}=0$ *exists,* $x=0$ is a removable discontinuity.

4. Find the values of a and b that make f continuous everywhere.

$$f(x)=\begin{cases}(x^2-4)/(x-2) & \text{if } x<2\\ ax^2-bx+3 & \text{if } 2\le x<3\\ 2x-a+b & \text{if } x\ge3.\end{cases}$$

Solution: Since $f(x)$ is continuous at $x=2$,

$\lim_{x\to2^-}f(x)=f(2)$. That is, $\lim_{x\to2^-}\dfrac{x^2-4}{x-2}=4=4a-2b+3$.

Since $f(x)$ is continuous at $x=3$, $\lim_{x\to3^-}f(x)=f(3)$.

That is, $\lim_{x\to3^-}ax^2-bx+3=3^2a-3b+3=2\cdot3-a+b$.

Therefore, $a=b=1/2$.

5. Use the intermediate value theorem to prove that $x^3+3x-2=0$ has a real solution between 0 and 1.

Proof: Let $f(x)=x^3+3x-2$. Then $f(x)$ is continuous on $[0,1]$. Since $f(0)=-2<0$ and $f(1)=2>0$, by the Intermediate Value Theorem, there is at least one number c between 0 and 1 such that $c^3+3c-2=0$.

6. Prove the equation $x=a+b\sin x$ (where $a>0$, $b>0$) has at least a positive root which does not exceed $a+b$.

Proof: Let $f(x)=a+b\sin(x)-x$. Then $f(x)$ is continuous on $[0,a+b]$, and $f(0)=a>0$,

$f(a+b)=b\sin(a+b)-b=b[\sin(a+b)-1]\le0$.

If $f(a+b)=b\sin(a+b)-b=0$, then $a+b$ is the positive root of the equation which does not exceeds $a+b$.

If $f(a+b)=b\sin(a+b)-b<0$, then since $f(0)=a>0$, we conclude by the Intermediate Value Theorem that $f(x)$ has at least one root in $(0,a+b)$.

Therefore, the equation $x=a+b\sin x$ (where $a>0$, $b>0$) has at least a positive root which does not exceed $a+b$.

7. Let $f(x+y)=f(x)+f(y)$ for all x and y and suppose that f is continuous at $x=0$. Prove that f is continuous everywhere.

Proof: Since $f(x)=f(x+0)=f(x)+f(0)$, $f(0)=0$.

Since f is continuous at $x=0$, $\lim_{h\to0}f(h)=f(0)=0$.

Then

$$\lim_{x\to c}f(x)=\lim_{h\to0}f(c+h)=\lim_{h\to0}[f(c)+f(h)]$$
$$=f(c)+\lim_{h\to0}f(h)=f(c)+0=f(c).$$

Hence f is continuous at c. Thus, f is continuous everywhere, since c is arbitrary.

Exercises 2.8 Comparisons between Infinitesimals

1. Suppose that as $x \to 0$, $x^2 \ln\left(1+x^2\right)$ is an infinitesimal of higher order than $x\sin x^n$, and $x\sin x^n$ is an infinitesimal of higher order than $1-\cos x$. Then the positive integer n = ___2___.

2. What is the equivalent infinitesimal of each of the following functions as $x \to 0$?

(1) $\sqrt[3]{1+\sqrt[3]{x}}-1$

(A) $x^{\frac{1}{3}}$ (B) $x^{\frac{1}{9}}$ (C) $\frac{1}{3}x^{\frac{1}{3}}$ (D) $\frac{1}{9}x^{\frac{1}{9}}$ [C]

(2) $\sqrt{1+\tan x}-\sqrt{1-\sin x}$;

(A) x (B) $2x$ (C) $x^{\frac{1}{2}}$ (D) $2x^{\frac{1}{2}}$ [A]

3. Find the following limits:

(1) $\lim\limits_{x\to 0}\frac{\tan 3x}{2x}$

Solution: As $x \to 0$, $\tan 3x \sim 3x$.

So, $\lim\limits_{x\to 0}\frac{\tan 3x}{2x}=\lim\limits_{x\to 0}\frac{3x}{2x}=\frac{3}{2}$.

(2) $\lim\limits_{x\to 1}\frac{\arcsin(1-x)}{\ln x}$; (Hint: let $1-x=t$)

Solution: Let $1-x=t$. So, $x=1-t$ and $t\to 0$ as $x\to 1$. As $t\to 0$, $\arcsin t \sim t$.

$$\lim_{x\to 1}\frac{\arcsin(1-x)}{\ln x}=\lim_{t\to 0}\frac{\arcsin t}{\ln(1-t)}=\lim_{t\to 0}\frac{t}{-t}=-1$$

(3) $\lim\limits_{x\to 1}\frac{1+\cos\pi x}{(x-1)^2}$; (Hint: let $x-1=t$)

Solution: Let $x-1=t$. So, $x=1+t$ and $t\to 0$ as $x\to 1$.

$$\lim_{x\to 1}\frac{1+\cos\pi x}{(x-1)^2}=\lim_{t\to 0}\frac{1+\cos\pi(1+t)}{t^2}=\lim_{t\to 0}\frac{1-\cos\pi t}{t^2}$$

$$=\lim_{t\to 0}\frac{\frac{(\pi t)^2}{2}}{t^2} \quad \left(t\to 0, 1-\cos\pi t\sim\frac{(\pi t)^2}{2}\right)$$

$$=\frac{\pi^2}{2}$$

(4) $\lim\limits_{x\to 0}\frac{\sqrt{1+x^3}-1}{\sin^3 x}$

Solution: As $x\to 0$, $\sin x\sim x$ and $\sqrt{1+x^3}-1\sim\frac{1}{2}x^3$.

Therefore, $\lim\limits_{x\to 0}\frac{\sqrt{1+x^3}-1}{\sin^3 x}=\lim\limits_{x\to 0}\frac{\frac{1}{2}x^3}{x^3}=\frac{1}{2}$.

(5) $\lim\limits_{x\to 0}\frac{e^x-1}{x}$

Solution: As $x\to 0$, $e^x-1\sim x$. Therefore,

$\lim\limits_{x\to 0}\frac{e^x-1}{x}=\lim\limits_{x\to 0}\frac{x}{x}=1$.

Exercises 3-1 Two Problems with One Theme

1. (1) Find the slope of the secant line to the curve $y = x^2 - 1$, passing through the two points $(0,-1)$ and $(1,0)$.

(2) Find the slope of the secant line to the curve $y = x^2 - 1$, passing through the two points $(1+h,(1+h)^2-1)$ and $(1,0)$.

(3) Find the slope of the tangent line to the curve $y = x^2 - 1$ at the point $x = 1$.

Solution: (1) The slope is $k_1 = \dfrac{-1-0}{0-1} = 1$.

(2) The slope is $k_2 = \dfrac{(1+h)^2-1-0}{1+h-1} = \dfrac{h^2+2h}{h} = h+2$.

(3) The slope is

$$k_3 = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{(1+h)^2-1-0}{h} = \lim_{h\to 0}\frac{h^2+2h}{h}$$
$$= \lim_{h\to 0}(h+2) = 2.$$

Exercises 3-2 The Derivative

1. Use the definition $f'(c) = \lim\limits_{h\to 0}\dfrac{f(c+h)-f(c)}{h}$ to find the derivative $f'(2)$ of $f(x) = x^2$.

Solution:

$$f'(2) = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h} = \lim_{h\to 0}\frac{(2+h)^2-2^2}{h} = \lim_{h\to 0}\frac{h^2+4h}{h}$$
$$= \lim_{h\to 0} h+4 = 4.$$

2. In the following problems, the given limit is a derivative, but of what function and at what point?

(1) $\lim\limits_{h\to 0}\dfrac{2(5+h)^3-2(5)^3}{h}$

Solution: The given limit is the derivative of $f(x) = 2x^3$ at $x = 5$.

.

(2) $\lim\limits_{x\to y}\dfrac{\sin x - \sin y}{x-y}$

Solution: The given limit is the derivative of $f(x) = \sin x$ at $x = y$.

3. Use the definition $f'(x) = \lim\limits_{h\to 0}\left[f(x+h)-f(x)\right]/h$ to find $f'\left(\dfrac{\pi}{3}\right)$ or prove $f'\left(\dfrac{\pi}{3}\right)$ does not exist.

$$f(x) = \begin{cases} \dfrac{3}{4\pi}x^2 + \dfrac{\sqrt{3}}{2} & \text{for } x < \dfrac{\pi}{3} \\ \dfrac{\pi}{12} + \sin x & \text{for } x \ge \dfrac{\pi}{3} \end{cases}$$

Solution: $f\left(\dfrac{\pi}{3}\right) = \dfrac{\pi}{12} + \sin\dfrac{\pi}{3} = \dfrac{\pi}{12} + \dfrac{\sqrt{3}}{2}$

$$\lim_{h\to 0^+}\frac{f\left(\frac{\pi}{3}+h\right)-f\left(\frac{\pi}{3}\right)}{h} = \lim_{h\to 0^+}\frac{\frac{\pi}{12}+\sin\left(\frac{\pi}{3}+h\right)-\frac{\pi}{12}-\frac{\sqrt{3}}{2}}{h}$$
$$= \lim_{h\to 0^+}\frac{\sin\left(\frac{\pi}{3}+h\right)-\sin\left(\frac{\pi}{3}\right)}{h}$$
$$= \lim_{h\to 0^+}\frac{2\sin\left(\frac{1}{2}\left(\frac{\pi}{3}+h-\frac{\pi}{3}\right)\right)\cos\left(\frac{1}{2}\left(\frac{\pi}{3}+h+\frac{\pi}{3}\right)\right)}{h}$$
$$= \lim_{h\to 0^+}\frac{\sin\left(\frac{h}{2}\right)}{h/2}\cos\left(\frac{\pi}{3}+\frac{h}{2}\right) = 1\times\cos\left(\frac{\pi}{3}+0\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\lim_{h\to 0^-}\frac{f\left(\frac{\pi}{3}+h\right)-f\left(\frac{\pi}{3}\right)}{h} = \lim_{h\to 0^-}\frac{\frac{3}{4\pi}\left(\frac{\pi}{3}+h\right)^2+\frac{\sqrt{3}}{2}-\frac{\pi}{12}-\frac{\sqrt{3}}{2}}{h}$$
$$= \frac{3}{4\pi}\lim_{h\to 0^-}\frac{\left(\frac{\pi}{3}+h\right)^2-\left(\frac{\pi}{3}\right)^2}{h}$$
$$= \frac{3}{4\pi}\lim_{h\to 0^-}\frac{\left(\frac{\pi}{3}+h-\frac{\pi}{3}\right)\left(\frac{\pi}{3}+h+\frac{\pi}{3}\right)}{h}$$
$$= \frac{3}{4\pi}\lim_{h\to 0^-}\left(\frac{2\pi}{3}+h\right) = \frac{3}{4\pi}\times\frac{2\pi}{3} = \frac{1}{2}$$

Since

$$\lim_{h\to 0^+}\frac{f\left(\frac{\pi}{3}+h\right)-f\left(\frac{\pi}{3}\right)}{h} = \frac{1}{2} = \lim_{h\to 0^-}\frac{f\left(\frac{\pi}{3}+h\right)-f\left(\frac{\pi}{3}\right)}{h}$$

$$f'\left(\frac{\pi}{3}\right) = \frac{1}{2}.$$

4. Suppose that $f(x+y) = f(x)f(y)$ for all x and y. Show that if $f'(0)$ exists then $f'(a)$ exists and $f'(a) = f(a)f'(0)$.

Proof: Since

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} = \lim_{h\to 0}\frac{f(a+h)-f(a+0)}{h}$$
$$= \lim_{h\to 0}\frac{f(a)f(h)-f(a)f(0)}{h}$$
$$= f(a)\lim_{h\to 0}\frac{f(h)-f(0)}{h} = f(a)f'(0),$$

$f'(a) = \lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$ exists and $f'(a) = f(a)f'(0)$.

5. Let $f(x) = \begin{cases} mx+b & \text{if } x<2 \\ x^2 & \text{if } x \ge 2 \end{cases}$. Determine m and b so that f is differentiable everywhere. (Hint: Thm A on Page 128)

Solution: If $f(x)$ is differentiable everywhere, then it is continuous everywhere. So, $\lim\limits_{x\to 2^-} f(x) = \lim\limits_{x\to 2^-}(mx+b) = 2m+b = f(2) = 4$, or $b = 4-2m$. Since $f(x)$ is differentiable at $x = 2$, $f'(2)$ exists. Since

$$\lim_{x\to 2^+}\frac{f(x)-f(2)}{x-2} = \lim_{x\to 2^+}\frac{x^2-4}{x-2} = \lim_{x\to 2^+}(x+2) = 4 \text{ and}$$

$$\lim_{x\to 2^-}\frac{f(x)-f(2)}{x-2} = \lim_{x\to 2^-}\frac{(mx+b)-4}{x-2} = \lim_{x\to 2^-}\frac{mx+(4-2m)-4}{x-2}$$
$$= \lim_{x\to 2^-}\frac{m(x-2)}{x-2} = m,$$

$m = 4$ and $b = 4-2(4) = -4$.

Exercises 3-3 Rules for Finding Derivatives

1. Find the derivatives of the following functions:

(1) $y=\pi x^3$;

Solution:

$D_x(\pi x^3)=\pi D_x(x^3)=\pi\cdot 3x^2=3\pi x^2$.

(2) $y=\dfrac{\alpha}{x^3}$;

Solution:

$D_x(\dfrac{\alpha}{x^3})=\alpha D_x(x^{-3})=\alpha(-3x^{-4})=-3\alpha x^{-4}=-\dfrac{3\alpha}{x^4}$

(3) $y=\left(x^2+2\right)\left(x^3+1\right)$;

Solution:

$D_x\left[\left(x^2+2\right)\left(x^3+1\right)\right]$

$=\left(x^2+2\right)D_x\left(x^3+1\right)+\left(x^3+1\right)D_x\left(x^2+2\right)$

$=\left(x^2+2\right)\left(3x^2\right)+\left(x^3+1\right)(2x)=5x^4+6x^2+2x$

(4) $y=\dfrac{x^2-x+1}{x^2+1}$.

Solution: $D_x\left[\dfrac{x^2-x+1}{x^2+1}\right]$

$=\dfrac{\left(x^2+1\right)D_x\left(x^2-x+1\right)-\left(x^2-x+1\right)D_x\left(x^2+1\right)}{\left(x^2+1\right)^2}$

$=\dfrac{\left(x^2+1\right)(2x-1)-\left(x^2-x+1\right)(2x)}{\left(x^2+1\right)^2}=\dfrac{x^2-1}{\left(x^2+1\right)^2}$

2. Find the equation of the tangent line to the curve $y=1/(x^2+4)$ at the point $(1,1/5)$.

Solution:

$D_x\left[\dfrac{1}{x^2+4}\right]=\dfrac{\left(x^2+4\right)D_x(1)-(1)D_x\left(x^2+4\right)}{\left(x^2+4\right)^2}$

$=\dfrac{\left(x^2+4\right)(0)-(1)(2x)}{\left(x^2+4\right)^2}=-\dfrac{2x}{\left(x^2+4\right)^2}$

At $x=1$: $m_{\tan}=-\dfrac{2(1)}{\left(1^2+4\right)^2}=-\dfrac{2}{25}$.

So, the equation of the tangent line is

$y-\dfrac{1}{5}=-\dfrac{2}{25}(x-1)\Rightarrow y=-\dfrac{2}{25}x+\dfrac{7}{25}$.

Exercises 3-4 Derivatives of Trigonometric Functions

1. Find the derivatives of the following functions.

(1) $y=\sin x\cos x$;

Solution:

$D_x(\sin x\cos x)=\sin xD_x\left[\cos x\right]+\cos xD_x\left[\sin x\right]$

$=\sin x(-\sin x)+\cos x(\cos x)=\cos^2 x-\sin^2 x$.

(2) $y=\dfrac{\sin x}{x}$;

Solution:

$D_x\left[\dfrac{\sin x}{x}\right]=\dfrac{xD_x(\sin x)-\sin xD_x(x)}{x^2}=\dfrac{x\cos x-\sin x}{x^2}$.

(3) $y=\tan^2 x$.

Solution: $y=\tan^2 x=(\tan x)(\tan x)$

$D_x y=(\tan x)(\sec^2 x)+(\tan x)(\sec^2 x)=2\tan x\sec^2 x$.

2. Show that the curves $y=\sqrt{2}\sin x$ and $y=\sqrt{2}\cos x$ intersect at right angles at a certain point with $0<x<\pi/2$.

Solution:

The curves intersect when $\sqrt{2}\sin x=\sqrt{2}\cos x$,

$\sin x=\cos x$ at $x=\dfrac{\pi}{4}$ for $0<x<\dfrac{\pi}{2}$.

$D_x\left(\sqrt{2}\sin x\right)=\sqrt{2}\cos x$; $\sqrt{2}\cos\dfrac{\pi}{4}=1$

$D_x\left(\sqrt{2}\cos x\right)=-\sqrt{2}\sin x$; $-\sqrt{2}\sin\dfrac{\pi}{4}=-1$.

Since $1(-1)=-1$, the two curves intersect at right angle.

Exercises 3-5 The Chain Rule

1. Find the derivatives of the following functions:

(1) $y=\sin t\tan(t^2+1)$;

Solution:

$D_t y=(\sin t)\cdot\left[\sec^2(t^2+1)\right](2t)+\tan(t^2+1)\cdot\cos t$

$=2t\sin t\sec^2(t^2+1)+\cos t\tan(t^2+1)$.

(2) $y=\sin^4\left(x^2+3x\right)$;

Solution:

$D_x\left[\sin^4(x^2+3x)\right]=4\sin^3(x^2+3x)D_x\left(\sin(x^2+3x)\right)$

$=4\sin^3(x^2+3x)\cos(x^2+3x)D_x(x^2+3x)$

$=4\sin^3(x^2+3x)\cos(x^2+3x)(2x+3)$

$=4(2x+3)\sin^3(x^2+3x)\cos(x^2+3x)$

(3) $y=\cos^4\left(\sin\theta^2\right)$.

Solution:

$D_\theta\left[\cos^4(\sin\theta^2)\right]=4\cos^3(\sin\theta^2)D_\theta\left[\cos(\sin\theta^2)\right]$

$=4\cos^3(\sin\theta^2)\left[-\sin(\sin\theta^2)\right]D_\theta\left(\sin\theta^2\right)$

$=-4\cos^3(\sin\theta^2)\sin(\sin\theta^2)\cos\theta^2 D_\theta\left(\theta^2\right)$

$=-8\theta\cos^3(\sin\theta^2)\sin(\sin\theta^2)\cos\theta^2$

2. In the following problems, express the indicated derivative in terms of the function $F(x)$. Assume that F is Differentiable.

(1) $\dfrac{d}{dz}\left(\dfrac{1}{\left(F(z)\right)^2}\right)$;

Solution: $\dfrac{d}{dz}\left(\dfrac{1}{\left(F(z)\right)^2}\right)=-2\left(F(z)\right)^{-3}F'(z)$.

(2) $\dfrac{d}{dx}F(\cos x)$.

Solution:

$\dfrac{d}{dx}F(\cos x)=F'(\cos x)\dfrac{d}{dx}(\cos x)=-\sin xF'(\cos x)$.

3. Given that $f(1)=2$, $f'(1)=-1$, $g(1)=0$ and $g'(1)=1$, find $F'(1)$, where $F(x)=f(x)\cos(g(x))$.

Solution:

$F'(x)=-f(x)g'(x)\sin g(x)+f'(x)\cos g(x)$

$F'(1)=-f(1)g'(1)\sin g(1)+f'(1)\cos g(1)$

$=-2(1)\sin 0-1\cos 0=-1$.

Exercises 3-6 Higher-Order Derivatives

1. Find $\dfrac{d^2y}{dx^2}$ of the function $y=\sin\left(x^3\right)$;

Solution:

$\dfrac{dy}{dx}=3x^2\cos(x^3)$

$\dfrac{d^2y}{dx^2}=3x^2\left[-3x^2\sin(x^3)\right]+6x\cos(x^3)=-9x^4\sin(x^3)+6x\cos(x^3)$.

Exercises 3-7 Implicit Differentiation

1. Assuming that equation $xy+\sin(xy)=1$ defines a differentiable function of x, find dy/dx by implicit differentiation.

Solution:

$$xD_xy+y+\cos(xy)(xD_xy+y)=0$$
$$xD_xy+x\cos(xy)D_xy=-y-y\cos(xy)$$
$$D_xy=\frac{-y-y\cos(xy)}{x+x\cos(xy)}=-\frac{y}{x}.$$

2. Find all points on the curve $x^2y-xy^2=2$ where the tangent line is vertical, that is, where $dx/dy=0$.

Solution: $x^2+2xy\dfrac{dx}{dy}-2xy-y^2\dfrac{dx}{dy}=0$

$$\frac{dx}{dy}(2xy-y^2)=2xy-x^2\Rightarrow\frac{dx}{dy}=\frac{2xy-x^2}{2xy-y^2}$$

$\dfrac{2xy-x^2}{2xy-y^2}=0$ if $x(2y-x)=0$, which occurs when $x=0$ or $y=\dfrac{x}{2}$. There are no points on $x^2y-xy^2=2$ where $x=0$. If $y=\dfrac{x}{2}$, then

$$2=x^2\left(\frac{x}{2}\right)-x\left(\frac{x}{2}\right)^2=\frac{x^3}{2}-\frac{x^3}{4}=\frac{x^3}{4}$$

so $x=2,\ y=\dfrac{2}{2}=1.$

3. Find the equation of the tangent line to the curve $x^2y^2+4xy=12y$ at the point $(2,1)$.

Solution:

$$x^2(2y)y'+2xy^2+4xy'+4y=12y'$$
$$y'(2x^2y+4x-12)=-2xy^2-4y$$
$$y'=\frac{-2xy^2-4y}{2x^2y+4x-12}=\frac{-xy^2-2y}{x^2y+2x-6}$$

At $(2,1)$, $y'=-2$.

So, the equation of the tangent line to the curve is $y-1=-2(x-2)$.

Exercises 3-9 Derivatives of Exponential and Logarithmic Functions

1. Find the indicated derivatives:

(1) $D_x\ln\left(x^2+3x+\pi\right)$;

Solution: $D_x\ln(x^2+3x+\pi)$

$$=\frac{1}{x^2+3x+\pi}\cdot D_x(x^2+3x+\pi)=\frac{2x+3}{x^2+3x+\pi}.$$

(2) $D_x\ln\left(x-4\right)^3$;

Solution:

$$D_x\ln(x-4)^3=D_x3\ln(x-4)=3\cdot\frac{1}{x-4}\cdot D_x(x-4)=\frac{3}{x-4}.$$

(3) D_xe^{x+2};

Solution: $D_xe^{x+2}=e^{x+2}D_x(x+2)=e^{x+2}$.

(4) D_xe^{-1/x^2}.

Solution:

$$D_xe^{-\frac{1}{x^2}}=e^{-\frac{1}{x^2}}D_x\left(-\frac{1}{x^2}\right)=e^{-\frac{1}{x^2}}\cdot 2x^{-3}=\frac{2e^{-\frac{1}{x^2}}}{x^3}.$$

2. Find dy/dx of the following functions by logarithmic differentiation.

(1) $y=\left(x^2+1\right)^{\ln x}$;

Solution: $\ln y=\ln\left(x^2+1\right)^{\ln x}=(\ln x)\ln(x^2+1)$

$$\frac{1}{y}\frac{dy}{dx}=\frac{d}{dx}\left[(\ln x)\ln(x^2+1)\right]$$
$$=\frac{1}{x}\ln(x^2+1)+\ln x\cdot\frac{2x}{x^2+1}$$
$$\frac{dy}{dx}=\left(x^2+1\right)^{\ln x}\left[\frac{\ln(x^2+1)}{x}+\frac{2x\ln x}{x^2+1}\right].$$

(2) $y=\left(\ln x^2\right)^{2x+3}$;

Solution:

$$\ln y=\ln\left(\ln x^2\right)^{2x+3}=(2x+3)\ln\left(\ln x^2\right).$$

$$\frac{1}{y}\frac{dy}{dx}=\frac{d}{dx}\left[(2x+3)\ln\left(\ln x^2\right)\right]$$
$$=2\ln\left(\ln x^2\right)+(2x+3)\frac{1}{\ln x^2}\frac{1}{x^2}(2x)$$
$$\frac{dy}{dx}=\left(\ln x^2\right)^{2x+3}\left[2\ln\left(\ln x^2\right)+\frac{2x+3}{x\ln x}\right].$$

(3) $y=\dfrac{\left(x^2+3\right)^{2/3}\left(3x+2\right)^2}{\sqrt{x+1}}$.

Solution:

$$\ln y=\frac{2}{3}\ln(x^2+3)+2\ln(3x+2)-\frac{1}{2}\ln(x+1)$$
$$\frac{1}{y}\frac{dy}{dx}=\frac{2}{3}\cdot\frac{2x}{x^2+3}+\frac{2\cdot 3}{3x+2}-\frac{1}{2(x+1)}$$
$$\frac{dy}{dx}=\frac{\left(x^2+3\right)^{2/3}\left(3x+2\right)^2}{\sqrt{x+1}}\left[\frac{4x}{3(x^2+3)}+\frac{6}{3x+2}-\frac{1}{2(x+1)}\right]$$
$$=\frac{(3x+2)(51x^3+70x^2+97x+90)}{6\left(x^2+3\right)^{1/3}\left(x+1\right)^{3/2}}.$$

Exercises 3-10 Derivatives of Inverse Trigonometric Functions

1. Find the derivatives of the following functions:

(1) $y=\arcsin(2x^2)$

Solution:

$$\frac{d}{dx}\arcsin(2x^2)=\frac{1}{\sqrt{1-(2x^2)^2}}\cdot 4x=\frac{4x}{\sqrt{1-4x^4}}.$$

(2) $y=\arccos(e^x)$

Solution:

$$\frac{d}{dx}\arccos(e^x)=-\frac{1}{\sqrt{1-(e^x)^2}}\cdot e^x=-\frac{e^x}{\sqrt{1-e^{2x}}}.$$

(3) $y=x^3\arctan(e^x)$

Solution:

$$\frac{d}{dx}\left[x^3\arctan(e^x)\right]=x^3\cdot\frac{e^x}{1+(e^x)^2}+3x^2\arctan(e^x)$$
$$=x^2\left[\frac{xe^x}{1+e^{2x}}+3\arctan(e^x)\right].$$

Exercises 3-8 Related Rates

1. Each edge of a variable cube is increasing at a rate of 3 inches per second. How fast is the volume of the cube increasing when an edge is 12 inches long?

Solution:

$V = x^3; \frac{dx}{dt} = 3$

$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$

When $x = 12$, $\frac{dV}{dt} = 3(12)^2(3) = 1296 \text{ in.}^3/\text{s}$

2. Assuming that a soap bubble retains its spherical shape as it expands, how fast is its radius increasing when its radius is 3 inches if air is blown into it at a rate of 3 cubic inches per second?

Solution:

Let r be the radius of the soap bubble, then the volume of the bubble is

$V = \frac{4}{3}\pi r^3$.

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.

At $r = 3$, $\frac{dV}{dt} = 3$,

$\frac{dr}{dt} = \frac{1}{12\pi} \approx 0.027$ in./s.

Exercises 3-11 Differentials and Approximations

1. Find dy of the following functions.

(1) $y = x^2 + x - 3$

Solution:

$dy = (2x+1)dx$

(2) $y = (\sin x + \cos x)^3$

Solution:

$dy = 3(\sin x + \cos x)^2(\cos x - \sin x)dx$

2. Find the linear approximation to the given functions at the specified points.

(1) $f(x) = \sqrt{1-x^2}$ at $x = 0$.

Solution:

$f(x) = \sqrt{1-x^2} \Rightarrow f'(x) = \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{1-x^2}}$

$a = 0$

The linear approximation is $L(x) = 1 + 0(x-0) = 1$.

(2) $G(x) = x + \sin 2x$ at $x = \frac{\pi}{2}$.

Solution:

$G(x) = x + \sin 2x$

$G'(x) = 1 + 2\cos 2x$

$a = \frac{\pi}{2}$

The linear approximation is then

$L(x) = \frac{\pi}{2} + (-1)\left(x - \frac{\pi}{2}\right) = -x + \pi$.

3. Use differential to approximate the number $\sqrt[3]{26.91}$.

Solution:

Let $f(x) = \sqrt[3]{x}$, $f'(x) = \frac{1}{3}x^{-2/3}$.

When $x_0 = 27$, $f(x_0) = \sqrt[3]{27} = 3$, $f'(x_0) = \frac{1}{3}27^{-2/3} = \frac{1}{27}$

$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$

$\sqrt[3]{26.91} \approx \sqrt[3]{27} + \frac{1}{3}27^{-2/3}(26.91 - 27) \approx 2.9967$.

Exercises 4-1 Maxima and Minima

In Problems 1-3, Identify the critical points and find the maximum and minimum values on the given interval.

1. $h(x) = e^{-x^2}$; $I = [-1, 3]$.

Solution:

$h'(x) = -2xe^{x^2}$; $h'(x) = 0$ when $x = 0$, which is in the domain $[-1, 3]$. Thus, this will be a critical point along with the endpoints of the interval.

Critical points: $-1, 0, 3$.

$h(-1) = \frac{1}{e}$, $h(0) = 1$, $h(3) = \frac{1}{e^9}$.

$h_{\max} = h(0) = 1$; $h_{\min} = h(3) = \frac{1}{e^9}$.

2. $f(x)=\dfrac{x}{1+x^2}$; $I=[-1,4]$

Solution: $f'(x)=\dfrac{1-x^2}{(1+x^2)^2}$; $f'(x)=0$ when $x=-1,1$.

Critical points: $-1,1,4$.

$f(-1)=-\dfrac{1}{2}, f(1)=\dfrac{1}{2}; f(4)=\dfrac{4}{17}$.

Therefore, $f_{\max}=f(1)=\dfrac{1}{2}, f_{\min}=f(-1)=-\dfrac{1}{2}$.

3. $g(x)=\sqrt[3]{x}$; $I=[-1,27]$

Solution:

$g'(x)=\dfrac{1}{3x^{2/3}}$; $g'(x)$ does not exist when x=0.

Critical points: $-1,0,27$.

$g(-1)=-1, g(0)=0, g(27)=3$.

Therefore, $g_{\max}=g(27)=3; g_{\min}=g(-1)=-1$.

Exercises 4-2 Monotonicity and Concavity

1. Use the Monotonicity Theorem to find where the given function is increasing and where it is decreasing.

(a) $h(z)=\dfrac{z^4}{4}-\dfrac{4z^3}{6}$;

Solution:

$h'(z)=z^3-2z^2=z^2(z-2).h'(z)=0$ when $z=0$, or 2.

Split $(-\infty,\infty)$ into the intervals $(-\infty,0),(0,2),(2,\infty)$.

x	$(-\infty,0)$	0	$(0,2)$	2	$(2,\infty)$
$h'(z)$	<0	0	<0	0	>0
$h(z)$	decreasing		decreasing		increasing

Therefore, $h(z)$ is increasing on $[2,\infty)$, and decreasing on $(-\infty,2]$.

(b) $f(x)=\dfrac{e^{-x}}{x^2}$;

Solution:

$$f'(x)=\frac{x^2\cdot(-e^{-x})-e^{-x}\cdot 2x}{\left(x^2\right)^2}=\frac{-xe^{-x}(x+2)}{x^4}$$

$f'(x)$ does not exist when $x=0$, $f'(x)=0$ when $x=-2$.

Split $(-\infty,\infty)$ into the intervals $(-\infty,-2),(-2,0),(0,\infty)$.

x	$(-\infty,-2)$	-2	$(-2,0)$	0	$(0,\infty)$
$f'(x)$	<0	0	>0	Not exist	<0
$f(x)$	decreasing		increasing		decreasing

Therefore, $f(x)$ is increasing on $[-2,0)$, and decreasing on $(-\infty,-2]$ and $(0.\infty)$.

2. Use the Concavity Theorem to determine where the given function is concave up and where it is concave down.

(a) $f(x)=x^4+8x^3-2$;

Solution: $f'(x)=4x^3+24x^2$

$f''(x)=12x^2+48x=12x(x+4)$.

x	$(-\infty,-4)$	-4	$(-4,0)$	0	$(0,\infty)$
$f''(x)$	>0	0	<0	0	>0
$f(x)$	Concave up	-258	Concave down	-2	Concave up

(b) $F(x)=2x^2+\cos^2 x$;

Solution: $F'(x)=4x-2\cos x\sin x$,

$F''(x)=2\sin^2 x-2\cos^2 x+4=2+4\sin^2 x$.

Since $2+4\sin^2 x>0$ for all x, $F(x)$ is concave up for all x.

3. Use the Monotonicity Theorem to prove the inequality

$$\frac{x}{1+x}<\ln(1+x)<x \quad \text{for } x>0.$$

Proof: Let $f(x)=\ln(1+x)-\dfrac{x}{1+x}$, $x\ge 0$. Then

$f'(x)=\dfrac{1}{1+x}-\dfrac{1}{(1+x)^2}=\dfrac{x}{(1+x)^2}>0$, for all $x>0$.

By Monotonicity Theorem, f is increasing on $[0,\infty)$. So $f(x)>f(0)=0$, for all $x>0$, i.e.

$$\frac{x}{1+x}<\ln(1+x) \quad \text{for } x>0.$$

Similarly, let $g(x)=\ln(1+x)-x$, $x\ge 0$. Then

$g'(x)=\dfrac{1}{1+x}-1=-\dfrac{x}{1+x}<0$, for all $x>0$.

By Monotonicity Theorem, g is decreasing on $[0,\infty)$. So $g(x)>g(0)=0$, for all $x>0$, i.e.

$$\ln(1+x)<x \quad \text{for } x>0.$$

4. Find the largest term in the sequence $\left(\sqrt[n]{n}\right)_{n=1}^{\infty}$.

Solution: Let $f(x)=x^{\frac{1}{x}}$, $f'(x)=x^{\frac{1}{x}}\left(\dfrac{\ln x}{x}\right)'=x^{\frac{1}{x}}\dfrac{1-\ln x}{x^2}$.

Solving $f'(x)=0$ obtains the only stationary point $x=e$.

When $0<x<e$, $f'(x)>0$, when $x>e$, $f'(x)<0$, hence $x=e$ is the global maximum.

Since $2<e<3$, the second or third term is the largest term in the sequence.

Comparing the two terms, we have $\sqrt[2]{2}<\sqrt[3]{3}$. Hence, the largest term is the third term, $\sqrt[3]{3}$.

Exercises 4-3 Local Extrema and Extrema on Open Intervals

1. Find the critical points of the following functions and use the test of your choice to decide which critical points give a local maximum value and which give a local minimum value. What are these local maximum and minimum values?

(1) $f(x)=x^3-12x+\pi$,

Solution:

$f'(x)=3x^2-12=3(x-2)(x+2)$.Critical points: $-2,2$.

(a)

x	$(-\infty,-2)$	-2	$(-2,2)$	2	$(2,\infty)$
$f'(x)$	>0	0	<0	0	>0
$f(x)$	increasing	$16+\pi$	decreasing	$\pi-16$	increasing

(b) $f''(x)=6x$. $f''(-2)=-12<0, f''(2)=12>0$.

Therefore, $f(x)$ reaches the local minimum at $x=2$ with the local minimum value $f(2)=\pi-16$, and $f(x)$ reaches the local maximum at $x=-2$ with the local maximum value $f(-2)=16+\pi$.

(2) $f(x)=\frac{1}{2}x+\sin x,\ 0<x<2\pi$

Solution:

$f'(x)=\frac{1}{2}+\cos x; \frac{1}{2}+\cos x=0$ when $\cos x=-\frac{1}{2}$.Critical points: $\frac{2\pi}{3},\frac{4\pi}{3}$.

(a)

x	$(0,\frac{2\pi}{3})$	$\frac{2\pi}{3}$	$(\frac{2\pi}{3},\frac{4\pi}{3})$	$\frac{4\pi}{3}$	$(\frac{4\pi}{3},2\pi)$
$f'(x)$	>0	0	<0	0	>0
$f(x)$	increasing	$\frac{\pi}{3}+\frac{\sqrt{3}}{2}$	decreasing	$\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$	increasing

Local minimum: $f(\frac{4\pi}{3})=\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$. Local maximum:

$f(\frac{2\pi}{3})=\frac{\pi}{3}+\frac{\sqrt{3}}{2}$.

(b) $f''(x)=-\sin x$, $f''(\frac{2\pi}{3})=-\sin(\frac{2\pi}{3})<0$, $f''(\frac{4\pi}{3})=-\sin(\frac{4\pi}{3})>0$

Therefore, Local minimum: $f(\frac{4\pi}{3})=\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$. Local maximum: $f(\frac{2\pi}{3})=\frac{\pi}{3}+\frac{\sqrt{3}}{2}$.

(3) $f(x)=\frac{\ln(x+2)}{x+2}$.

Solution:

The domain of the function is $(-2,\infty)$. $f'(x)=\frac{1-\ln(x+2)}{(x+2)^2}$

Solving $f'(x)=0$, we have $x=e-2$.

The only critical points are stationary points. $e-2$.

(a) $f'(x)>0$ on $(-2,e-2)$, $f'(x)<0$ on $(e-2,\infty)$.

$f(x)$ reaches the local maximum $f(e-2)=\frac{1}{e}$ at $x=e-2$.

(b) $f''(x)=\frac{2\ln(x+2)-3}{(x+2)^3}$, and $f''(e-2)=-\frac{1}{e^3}<0$.

By the second derivative test,

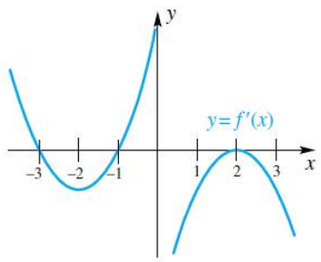
$f(x)$ reaches the local maximum at $x=e-2$.

2. Let $f(x)$ be a continuous function and let $f'(x)$ have the graph shown in the following figure. Answer the following questions.

(a) Where is f increasing ? Decreasing?

(b) Where is f concave up? Concave down?

(c) Where does f attain a local maximum? A local minimum?



Solution:

(a) f is increasing on $(-\infty,-3]$ and $[-1,0)$, and decreasing on $[-3,-1]$ and $(0,\infty)$.

(b) f is concave up on $(-2,0)$ and $(0,2)$, concave down on $(-\infty,-2)$ and $(2,\infty)$.

(c) f attains local maxima at $x=-3$ and $x=0$, and a local minimum at $x=-1$.

3. Find, if possible, the (global) maximum and minimum values of the function $f(x)=6\sqrt{x}-4x$ on $[0,4]$.

Solution:

$f'(x)=\frac{3}{\sqrt{x}}-4=\frac{3-4\sqrt{x}}{\sqrt{x}}$.

$f'(x)$ is not well-defined when $x=0$, and $f'(x)=0$ when $x=\frac{9}{16}$.

Critical points: $0,\frac{9}{16},4$.

$f(0)=0, f(\frac{9}{16})=\frac{9}{4}, f(4)=-4$.

Therefore, $f_{\max}=f(\frac{9}{16})=\frac{9}{4}, f_{\min}=f(4)=-4$.

4. For what values of the numbers a and b does the function $f(x)=axe^{bx^2}$ have the maximum value $f(2)=1$?

Solution: $f'(x)=a(1+2bx^2)e^{bx^2}$

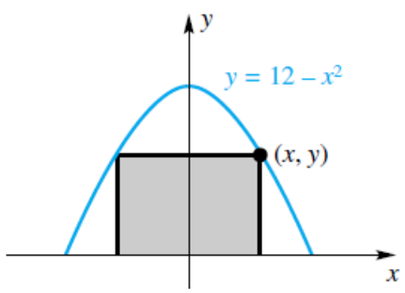
Since $f(x)$ have the maximum value $f(2)=1$, we have

$$\begin{cases} f'(2)=0 \\ f(2)=1 \end{cases} \Rightarrow \begin{cases} a(1+8b)=0 \\ 2ae^{4b}=1 \end{cases} \Rightarrow \begin{cases} a=\frac{\sqrt{e}}{2} \\ b=-\frac{1}{8} \end{cases}.$$

When $a=\frac{\sqrt{e}}{2}$ and $b=-\frac{1}{8}$, $f''(2)<0$.

Exercises 4-4 Practical Problems

1. A rectangle has two corners on the x-axis and the other two on the parabola $y = 12 - x^2$, with $y \geq 0$. What are the dimensions of the rectangle of this type with maximum area?



Solution:

According to the assumption, $x \in \left[0, \sqrt{12}\right]$.

$A = 2x\left(12 - x^2\right) = 24x - 2x^3$. $\frac{\mathrm{d}A}{\mathrm{d}x} = 24 - 6x^2$.

$\frac{\mathrm{d}A}{\mathrm{d}x} = 0$ when $x = -2, 2$.

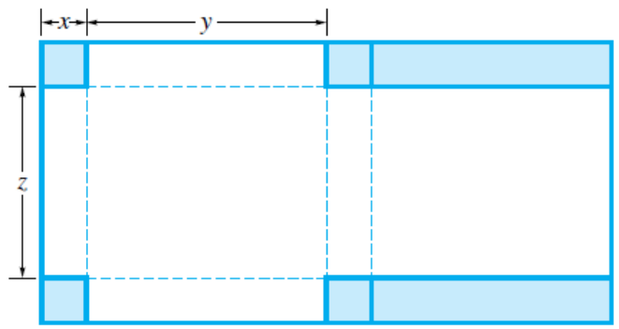
Critical points: $0, 2, \sqrt{12}$.

When $x = 0$ or $\sqrt{12}$, $A = 0$.

When $x = 2$, $y = 12 - (2)^2 = 8$, $2x = 4$.

The dimensions are 4 by 8 (i.e. $x = 2, y = 8$).

2. A covered box is to be made from a rectangular sheet of cardboard measuring 5 feet by 8 feet. This is done by cutting out the shaded regions of Figure and then folding on the dotted lines. What are the dimensions x, y, and z that maximize the volume?



Solution:

From the given graph,

$$\begin{cases} 2x + 2y = 8 \\ 2x + z = 5 \end{cases} \Rightarrow \begin{cases} y = 4 - x \\ z = 5 - 2x \end{cases}$$

And the following three inequalities for x should hold

$x \geq 0, 4 - x \geq 0$ and $5 - 2x \geq 0$. Thus $0 \leq x \leq \frac{5}{2}$.

$V = xyz = x(4 - x)(5 - 2x) = 20x - 13x^2 + 2x^3$.

$\frac{\mathrm{d}V}{\mathrm{d}x} = 20 - 26x + 6x^2 = 2(3x - 10)(x - 1)$.

$\frac{\mathrm{d}V}{\mathrm{d}x} = 0$ when $x = 1, \frac{10}{3}$.

Critical points: $0, 1, \frac{5}{2}$.

At $x = 0$, or $\frac{5}{2}$, $V = 0$. At $x = 1$, $V = 9$.

When $x = 1, y = 3, z = 3$, the volume reaches the maximum 9.

3. A cylindrical can is to be made to hold 1L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Solution:

Let r be the radius of the base and h the height (both in cm).

To minimize the cost of the metal, we minimize the total surface are of the can, which is

$$A = 2\pi r^2 + 2\pi rh$$

The volume of the can is 1L (=1000cm^3), which means

$$\pi r^2 h = 1000 \Rightarrow h = 1000 / (\pi r^2).$$

Then

$$A = 2\pi r^2 + 2\pi rh = 2\pi r^2 + 2\pi r \frac{2000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}$$

$\frac{\mathrm{d}A}{\mathrm{d}r} = 4\pi r - \frac{2000}{r^2} = 0 \Rightarrow r = \sqrt[3]{\frac{500}{\pi}}$ is the only critical point.

Because $\frac{\mathrm{d}A}{\mathrm{d}r}$ <0 on $0 < r < \sqrt[3]{\frac{500}{\pi}}$ and $\frac{\mathrm{d}A}{\mathrm{d}r}$ >0 on $\sqrt[3]{\frac{500}{\pi}} < r < \infty$, $A(r)$ attains local minimum at $r = \sqrt[3]{\frac{500}{\pi}}$, which is also the global minimum.

So when $r = \sqrt[3]{\frac{500}{\pi}}$ and $h = \frac{1000}{\pi r^2} = 2r$, the cost of metal is minimized.

Exercises 4-6 Mean Value Theorems

1. Show that $f(x)=2x^3-9x^2+1=0$ has exactly one solution on the interval (0, 1).

Solution:

$f'(x)=6x^2-18x=6x(x-3)$;

$f'(x)=0$ when $x=0$ or $x=3$.

$f(0)=1, f(1)=-6$, , by the Intermediate Value Theorem, there is at least one point c on (0, 1) with $f(c)=0$.

Suppose there are two points, c and $c', c<c'$ on (0,1) with $f(c)=f(c')=0$. Then by Rolle's Theorem, there is at least one value d on $(c,c')\subset(0,1)$ with $f'(d)=0$. This contradicts with the fact that $f'(x)\neq 0$ for all x on (0,1), thus there cannot be more than one x on (0,1) where $f(x)=0$, i.e., $f(x)=0$ has exactly one solution on (0,1).

2. Let $f(x)=(x-1)(x-2)(x-3)$. Prove that there is at least one value in the interval [0, 4] where $f''(x)=0$ and two values in the same interval where $f'(x)=0$.

Proof: $f(x)$ is a polynomial function so it is continuous on [0, 4] and $f''(x)$ exists for all x in (0, 4).

$f(1)=f(2)=f(3)=0$, so by Rolle's Theorem, there are at least two values $x_1\in(1,\ 2)\subset(0,4),\ x_2\in(2,\ 3)\subset(0,4)$ such that $f'(x_1)=f'(x_2)=0$.

Moreover, by applying Rolle's Theorem to $f'(x)$, at least one value $x_3\in(x_1,x_2)\subset(0,4)$ where $f''(x_3)=0$.

3. Use the Mean Value Theorem to show the inequality

$$|\sin x-\sin y|\le|x-y| \text{ for all } x \text{ and } y.$$

Solution:

For $y=x$, the inequality becomes $0\le 0$, which is true.

If x and y are not the same, assume that $y<x$. Define $f(z)=\sin z$. Then $f'(z)=\cos z$ and $|f'(z)|\le 1$ for all $z\in R$.

By applying the Mean Value Theorem and taking the absolute value of both sides, we have

$$\frac{|f(x)-f(y)|}{|x-y|}=|f'(c)|, \text{ for a number } c\in(y,x)$$

Since $|f'(c)|\le 1$, we have $|\sin x-\sin y|\le|x-y|$.

Exercises 4-7 Indeterminate Forms of Type 0/0

Find the following indicated limits. Problem 1-Problem 8.

1. $\lim\limits_{x\to 0}\dfrac{x^3-3x^2+x}{x^3-2x}$

Solution: $\lim\limits_{x\to 0}\dfrac{x^3-3x^2+x}{x^3-2x}=\lim\limits_{x\to 0}\dfrac{3x^2-6x+1}{3x^2-2}=-\dfrac{1}{2}$

2. $\lim\limits_{x\to 1}\dfrac{\ln x^2}{x^2-1}$

Solution: $\lim\limits_{x\to 1}\dfrac{\ln x^2}{x^2-1}=\lim\limits_{x\to 1}\dfrac{\frac{1}{x^2}\cdot 2x}{2x}=\lim\limits_{x\to 1}\dfrac{1}{x^2}=1$

3. $\lim\limits_{x\to 0}\dfrac{e^x-e^{-x}}{2\sin x}$

Solution: $\lim\limits_{x\to 0}\dfrac{e^x-e^{-x}}{2\sin x}=\lim\limits_{x\to 0}\dfrac{e^x+e^{-x}}{2\cos x}=\dfrac{2}{2}=1$

4. $\lim\limits_{t\to 1}\dfrac{\sqrt{t}-t^2}{\ln t}$

Solution: $\lim\limits_{t\to 1}\dfrac{\sqrt{t}-t^2}{\ln t}=\lim\limits_{t\to 1}\dfrac{\frac{1}{2\sqrt{t}}-2t}{\frac{1}{t}}=\dfrac{\frac{1}{2}-2}{1}=-\dfrac{3}{2}$

5. $\lim\limits_{x\to 0}\dfrac{\tan x-x}{\sin 2x-2x}$

Solution:

$$\lim_{x\to 0}\frac{\tan x-x}{\sin 2x-2x}=\lim_{x\to 0}\frac{\sec^2 x-1}{2\cos 2x-2}=\lim_{x\to 0}\frac{\tan^2 x}{-4\sin^2 x}$$
$$=-\frac{1}{4}\lim_{x\to 0}\frac{1}{\cos^2 x}=-\frac{1}{4}.$$

6. $\lim\limits_{x\to 0^+}\dfrac{x^2}{\sin x-x}$

Solution: $\lim\limits_{x\to 0^+}\dfrac{x^2}{\sin x-x}=\lim\limits_{x\to 0^+}\dfrac{2x}{\cos x-1}=\lim\limits_{x\to 0^+}\dfrac{2}{-\sin x}$

As $x\to 0^+, -\sin x\to 0^-$, so $\lim\limits_{x\to 0^+}\dfrac{2}{-\sin x}=-\infty$

or $\lim\limits_{x\to 0^+}\dfrac{x^2}{\sin x-x}$ does not exist.

7. $\lim\limits_{x\to 0}\dfrac{e^x-\ln(1+x)-1}{x^2}$

Solution: $\lim\limits_{x\to 0}\dfrac{e^x-\ln(1+x)-1}{x^2}=\lim\limits_{x\to 0}\dfrac{e^x-\frac{1}{1+x}}{2x}=\lim\limits_{x\to 0}\dfrac{e^x+\frac{1}{(1+x)^2}}{2}=1$

8. $\lim\limits_{x\to 0}\dfrac{\arctan x-x}{8x^3}$

Solution:

$$\lim_{x\to 0}\frac{\tan^{-1}x-x}{8x^3}=\lim_{x\to 0}\frac{\frac{1}{1+x^2}-1}{24x^2}=\lim_{x\to 0}\frac{\frac{-2x}{(1+x^2)^2}}{48x}=-\lim_{x\to 0}\frac{1}{24(1+x^2)^2}=-\frac{1}{24}$$

Exercises 4-8 Other Indeterminate Forms

Find the following indicated limits. Problem 1-Problem 9.

1. $\lim\limits_{x\to\infty}\dfrac{x^2}{e^x}$

Solution: $\lim\limits_{x\to\infty}\dfrac{x^2}{e^x}=\lim\limits_{x\to\infty}\dfrac{2x}{e^x}=\lim\limits_{x\to\infty}\dfrac{2}{e^x}=0$

2. $\lim\limits_{x\to\pi/2}\dfrac{3\sec x+5}{\tan x}$

Solution:

$$\lim_{x\to\pi/2}\frac{3\sec x+5}{\tan x}=\lim_{x\to\pi/2}\frac{3\sec x\tan x}{\sec^2 x}=\lim_{x\to\pi/2}\frac{3\tan x}{\sec x}$$

$$=\lim_{x\to\pi/2}3\sin x=3$$

3. . $\lim\limits_{x\to 0^+}x^2\ln(\sin x)$

Solution:

$$\lim_{x\to 0^+}x^2\ln(\sin x)=\lim_{x\to 0^+}\frac{\ln(\sin x)}{\frac{1}{x^2}}$$

$$=\lim_{x\to 0^+}\frac{\frac{\cos x}{\sin x}}{-\frac{2}{x^3}}=\lim_{x\to 0^+}\left(-\frac{x^3\cos x}{2\sin x}\right)$$

$$=\lim_{x\to 0^+}\left(-\frac{x}{\sin x}\cdot\frac{x^2\cos x}{2}\right)=-1\cdot 0=0$$

4. $\lim\limits_{x\to 0}(\cos x)^{\sin x}$

Solution:

$$\lim_{x\to 0}(\cos x)^{\sin x}=e^{\lim\limits_{x\to 0}\sin x\ln(\cos x)}=e^0=1$$

5. $\lim\limits_{x\to 1}(\dfrac{1}{x-1}-\dfrac{x}{\ln x})$

Solution:

$$\lim_{x\to 1}(\frac{1}{x-1}-\frac{x}{\ln x})=\lim_{x\to 1}\frac{\ln x-x^2+x}{(x-1)\ln x}$$

$$=\lim_{x\to 1}\frac{\frac{1}{x}-2x+1}{\ln x+\frac{x-1}{x}}=\lim_{x\to 1}\frac{1-2x^2+x}{x\ln x+x-1}$$

$$=\lim_{x\to 1}\frac{-4x+1}{\ln x+2}=-\frac{3}{2}$$

6. $\lim\limits_{x\to\pi/2}(\tan x-\sec x)$

Solution:

$$\lim_{x\to\pi/2}(\tan x-\sec x)=\lim_{x\to\pi/2}\frac{\sin x-1}{\cos x}=\lim_{x\to\pi/2}\frac{\cos x}{-\sin x}=0$$

7. $\lim\limits_{x\to 0^+}(3x)^{x^2}$

Solution:

Let $y=(3x)^{x^2}$ then $\ln y=x^2\ln 3x$

$$\lim_{x\to 0^+}x^2\ln 3x=\lim_{x\to 0^+}\frac{\ln 3x}{\frac{1}{x^2}}=\lim_{x\to 0^+}\frac{\frac{1}{3x}\cdot 3}{-2\frac{1}{x^3}}$$

$$=-\lim_{x\to 0^+}\frac{x^2}{2}=0$$

$$\lim_{x\to 0^+}(3x)^{x^2}=\lim_{x\to 0^+}e^{\ln y}=e^{\lim\limits_{x\to 0^+}\ln y}=e^0=1$$

8. $\lim\limits_{x\to\infty}x^{1/x}$

Solution:

Let $y=x^{1/x}$ then $\ln y=\dfrac{1}{x}\ln x$

$$\lim_{x\to\infty}\frac{1}{x}\ln x=\lim_{x\to\infty}\frac{\ln x}{x}=\lim_{x\to\infty}\frac{1}{x}=0$$

$$\lim_{x\to\infty}x^{1/x}=\lim_{x\to\infty}e^{\ln y}=e^0=1$$

9. $\lim\limits_{x\to 0}(\cos x)^{1/x^2}$

Solution: Let $y=(\cos x)^{1/x^2}$ then $\ln y=\dfrac{1}{x^2}\ln\cos x$

$$\lim_{x\to 0}\frac{1}{x^2}\ln\cos x=\lim_{x\to 0}\frac{\ln\cos x}{x^2}=\lim_{x\to 0}\frac{-\frac{\sin x}{\cos x}}{2x}$$

$$=\lim_{x\to 0}\frac{-\tan x}{2x}=\lim_{x\to 0}\frac{-\sec^2 x}{2}=-\frac{1}{2}$$

$$\lim_{x\to 0}(\cos x)^{1/x^2}=\lim_{x\to 0}e^{\ln y}=e^{-\frac{1}{2}}$$

10. First verify that l'Hôpital's Rule is not applicable to the limit $\lim\limits_{x\to\infty}\dfrac{x+\sin x}{x}$. Then use other method to find the limit.

Solution: Let $f(x)=x+\sin x,\ g(x)=x$. Then

$\lim\limits_{x\to\infty}\dfrac{f'(x)}{g'(x)}=\lim\limits_{x\to\infty}(1+\cos x)$ does not exist and also do not approaches infinity. So we cannot use the l'Hôpital's Rule.

$$\lim_{x\to\infty}\frac{x+\sin x}{x}=\lim_{x\to\infty}\frac{x}{x}+\lim_{x\to\infty}\frac{\sin x}{x}=1+0=1.$$

Exercises 4-10 Antiderivatives

Find the general antiderivatives for Problem 1- Problem 6.

1. $f(x) = x - 4$

Solution:

$$\int (x-4)\,dx = \int x dx - 4\int dx = \frac{x^2}{2} - 4x + C$$

2. $f(x) = 3x^2 + \sqrt{3}$

Solution:

$$\int \left(3x^2 + \sqrt{3}\right)dx = 3\int x^2 dx + \sqrt{3}\int dx = x^3 + \sqrt{3}x + C$$

3. $f(x) = 7x^{-3/4}$

Solution:

$$\int 7x^{-3/4}dx = 7\int x^{-3/4}dx = 7\left(4x^{1/4} + C_1\right)$$
$$= 28x^{1/4} + C$$

4. $f(x) = \dfrac{\sqrt{2x}}{x} + \dfrac{3}{x^5}$

Solution:

$$\int \left(\frac{\sqrt{2x}}{x} + \frac{3}{x^5}\right)dx = \sqrt{2}\int \frac{1}{\sqrt{x}}dx + 3\int x^{-5}dx$$
$$= 2\sqrt{2}\sqrt{x} - \frac{3}{4}x^{-4} + C$$

5. $f(x) = \dfrac{x^6 - x}{x^3}$

Solution:

$$\int \left(\frac{x^6 - x}{x^3}\right)dx = \int x^3 dx - \int x^{-2}dx = \frac{1}{4}x^4 + x^{-1} + C$$

6. $f(x) = x^3 + e^x$

Solution:

$$\int \left(x^3 + e^x\right)dx = \int x^3 dx + \int e^x dx = \frac{1}{4}x^4 + e^x + C$$

Evaluate the indicated indefinite integrals for 7-9.

7. $\int (x^3 + \sqrt{x})dx$

Solution:

$$\int \left(x^3 + \sqrt{x}\right)dx = \int x^3 dx + \int x^{1/2}dx$$
$$= \frac{x^4}{4} + \frac{x^{3/2}}{\frac{3}{2}} + C = \frac{x^4}{4} + \frac{2}{3}x^{3/2} + C$$

8. $\int (z + \sqrt{2}z)^2 dz$

Solution:

$$\int \left(z + \sqrt{2}z\right)^2 dz = \int \left[\left(1+\sqrt{2}\right)z\right]^2 dz$$
$$= \left(1+\sqrt{2}\right)^2 \int z^2 dz = \frac{\left(1+\sqrt{2}\right)^2 z^3}{3} + C$$

9. $\int \dfrac{s(s+1)^2 ds}{\sqrt{s}}$

Solution:

$$\int \frac{s\left(s+1\right)^2}{\sqrt{s}}ds = \int \frac{s^3 + 2s^2 + s}{\sqrt{s}}ds$$
$$= \int s^{5/2}ds + 2\int s^{3/2}ds + \int s^{1/2}ds$$
$$= \frac{2s^{7/2}}{7} + \frac{4s^{5/2}}{5} + \frac{2s^{3/2}}{3} + C$$

In Problems 10–18, evaluate the indefinite integrals.

10. $\int (\sqrt{2}x+1)^3 dx$;

Solution: Let $g(x)=\sqrt{2}x+1$. Then $g'(x)=\sqrt{2}$.

$$\int \left(\sqrt{2}x+1\right)^3 dx = \frac{1}{\sqrt{2}}\int [g(x)]^3 g'(x)dx = \frac{1}{\sqrt{2}}\int [g(x)]^3 dg(x)$$

$$= \frac{[g(x)]^4}{4\sqrt{2}} + C = \frac{\left(\sqrt{2}x+1\right)^4}{4\sqrt{2}} + C$$

11. $\int x^2(\pi x^3+1)^4 dx$

Solution: Let $g(x)=\pi x^3+1$. Then $g'(x)=3\pi x^2$.

$$\int x^2\left(\pi x^3+1\right)^4 dx = \frac{1}{3\pi}\int [g(x)]^4 g'(x)dx = \frac{1}{3\pi}\int [g(x)]^4 dg(x)$$

$$= \frac{[g(x)]^5}{15\pi} + C = \frac{\left(\pi x^3+1\right)^5}{15\pi} + C$$

12. $\int (5x^2+1)\sqrt{5x^3+3x-2}dx$

Solution: Let $u=5x^3+3x-2$. Then $du=\left(15x^2+3\right)dx$.

$$\int \left(5x^2+1\right)\sqrt{5x^3+3x-2}dx = \int \frac{1}{3}\left(15x^2+3\right)\sqrt{5x^3+3x-2}dx$$

$$= \frac{1}{3}\int u^{1/2}du = \frac{1}{3}\left(\frac{2}{3}u^{3/2}+C_1\right) = \frac{2}{9}\left(5x^3+3x-2\right)^{3/2} + C$$

13. $\int \ln t dt$

Solution: Let $u=\ln t$ and $v=t$. Here $du=\frac{1}{t}dt$.

$$\int \ln t dt = \int u dv = uv - \int v du = t\ln t - \int t d\ln t$$

$$= t\ln t - \int \frac{t}{t}dt = t\ln t - t + C$$

14. $\int x\sin x dx$

Solution: $\int x\sin x dx = -\int x d\cos x$

Let $u=x$ and $v=\cos x$.

$$\int x\sin x dx = -\int x d\cos x = -\int u dv = -uv + \int v du$$

$$= -x\cos x + \int \cos x dx = -x\cos x + \sin x + C$$

15. $\int \sin x(1+\cos x)^4 dx$

Solution: Let $u=1+\cos x$. Then $du=-\sin x dx$.

$$\int \sin x\left(1+\cos x\right)^4 dx = -\int -\sin x\left(1+\cos x\right)^4 dx$$

$$= -\int u^4 du = -\left(\frac{1}{5}u^5 + C_1\right)$$

$$= -\frac{1}{5}\left(1+\cos x\right)^5 + C$$

16. $\int \sin x\cos x\sqrt{1+\sin^2 x}dx$

Solution: Let $u=1+\sin^2 x$. Then $du=2\sin x\cos x dx$.

$$\int \sin x\cos x\sqrt{1+\sin^2 x}dx = \int \frac{1}{2}\cdot 2\sin x\cos x\sqrt{1+\sin^2 x}dx$$

$$= \frac{1}{2}\int \sqrt{u}du = \frac{1}{2}\left(\frac{2}{3}u^{3/2}+C_1\right) = \frac{1}{3}\left(1+\sin^2 x\right)^{3/2} + C$$

17. $\int (1+e^x)^2 e^x dx$

Solution: Let $u=1+e^x$. Then $du=e^x dx$.

$$\int \left(1+e^x\right)^2 e^x dx = \int u^2 du = \frac{1}{3}u^3 + C$$

$$= \frac{1}{3}\left(1+e^x\right)^3 + C$$

18. $\int \frac{3y}{\sqrt{5+2y^2}}dy$

Solution: Let $u=2y^2+5$. Then $du=4ydy$

$$\int \frac{3y}{\sqrt{2y^2+5}}dy = \int \frac{3}{4}(4y)\left(2y^2+5\right)^{-1/2}dy$$

$$= \frac{3}{4}\int u^{-1/2}du = \frac{3}{4}\left(2u^{1/2}+C_1\right)$$

$$= \frac{3}{2}\sqrt{2y^2+5} + C$$

Exercises 4-11 Introductions to Differential Equations

1. Find the general solution for each differential equation.

(1) $\frac{dy}{dx}=3x^2y-2xy$;

Solution: Since $\frac{dy}{dx}=y\left(3x^2-2x\right)$, $\frac{dy}{y}=\left(3x^2-2x\right)dx$.

Integrating on both sides, we have $\int\frac{dy}{y}=\int\left(3x^2-2x\right)dx$.

That is, $\ln|y|=x^3-x^2+C$. Therefore,

$|y|=e^{\ln|y|}=e^{x^3-x^2+C}=e^Ce^{x^3-x^2}=he^{x^3-x^2}$. Here $h>0$. Then

$y=\pm h\left(e^{x^3-x^2}\right)$. That is, $y=ke^{x^3-x^2}, k\in\mathbf{R}$ and $k\neq 0$. If

$y\equiv 0$, $\frac{dy}{dx}=0=(3x^2-2x)\cdot 0$.

Hence, $y=ke^{x^3-x^2}, k\in\mathbf{R}$.

(2) $(y^2-y)\frac{dy}{dx}=x$.

Solution: Since $\left(y^2-y\right)dy=xdx$, $\int\left(y^2-y\right)dy=\int xdx$.

That is, $\frac{y^3}{3}-\frac{y^2}{2}=\frac{x^2}{2}+C$. So, $2y^3-3y^2=3x^2+C$.

2. Find the particular solution for each initial value problem

(1) $\frac{dy}{dx}=4x^3-3x^2+x;\quad y(1)=0$;

Solution: $y=\int\left(4x^3-3x^2+x\right)dx=x^4-x^3+\frac{x^2}{2}+C$. Let

$x=1$ and $y=0$, we have $0=1-1+\frac{1}{2}+C$. So, $C=-\frac{1}{2}$

and $y=x^4-x^3+\frac{x^2}{2}-\frac{1}{2}$.

(2) $\frac{dy}{dx}=\frac{x^2+5}{2y-1};\quad y(0)=11$.

Solution: Since $(2y-1)dy=\left(x^2+5\right)dx$,

$\int(2y-1)dy=\int\left(x^2+5\right)dx$.

That is, $y^2-y=\frac{x^3}{3}+5x+C$. Let $x=0$ and $y=11$, we

have $121-11=C$. That is, $C=110$. So, $y^2-y=\frac{x^3}{3}+5x+110$.

3. If an object moves along a coordinate line from the origin $x=0$ with an initial velocity $v_0=0$ (m/s) and its acceleration is $a(t)=\sin t$ (m/s^2), what is its position at the instant $t=\frac{\pi}{2}$ (s)?

Solution: $s''(t)=a(t)=\sin t\Rightarrow s'(t)=-\cos t+C_1$

$v_0=s'(0)=-1+C_1=0\Rightarrow C_1=1$

$s'(t)=-\cos t+1\Rightarrow s(t)=-\sin t+t+C_2$

$s(0)=C_2=0\Rightarrow s(t)=-\sin t+t\Rightarrow s(\frac{\pi}{2})=\frac{\pi}{2}-1$.

Exercises 4-12 Exponential Growth and Decay

1. A population is growing at a rate proportional to its size. After 5 years, the population size was 164000. After 12 years, the population size was 235000. What was the original population size?

Solution: Assume that the population size was $x(t)$ after *t* years. Then $\frac{dx}{dt}=kx$, where *k* is a constant.

The solution of the differential equation is $x=Ce^{kt}$, where $C=x(0)$ is the original population size. Then we have following two equations

$$\begin{cases}Ce^{5k}=164000\\ Ce^{12k}=235000\end{cases}\Rightarrow\begin{cases}k\approx 0.0514\\ C\approx 126840\end{cases}$$

The original population size was 126840.

2. Consider a country with a population of 10 million in 1985, a growth rate of 1.2% per year, and immigration from other countries of 60000 per year. Use differential equation

$$\frac{dy}{dt}=ay+b,\quad y(0)=y_0$$

to model this situation, where $a=1.2\%, b=0.06$ and $y_0=10$. Predict the population in 2010.

Solution: Assume that the population size is $x(t)$ in the *t*'th year after year 1985. The question is to find out $x(25)$, the population in 2010.

Then $\frac{dx}{dt}=0.012x+0.06\Rightarrow\frac{1}{0.012x+0.06}dx=dt$

$\Rightarrow\ \ln(0.012x+0.06)=0.012t+C_1$

$\Rightarrow\ 0.012x+0.06=C_2e^{0.012t}$

$\Rightarrow\ x(t)=Ce^{0.012t}-5$.

When $t=0$, $x(0)=10$, $10=Ce^{0.012\times 0}-5\ \Rightarrow C=15$

Hence, we have $x(t)=15e^{0.012t}-5$.

In 2010, the population will be

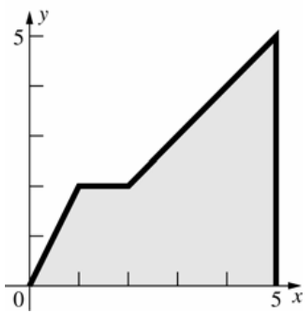
$x(25)=15e^{0.012\times 25}-5=15e^{0.3}-5\approx 15.25$ million.

Exercises 5.2 The Definite Integral

1. In the following problems, calculate $\int_a^b f(x)\,dx$ where a and b are the left and right end points for which $f(x)$ is defined. By using the Interval Additive Property and the appropriate area formulas from plane geometry. Begin by graphing the given function.

(a). $f(x)=\begin{cases} 2x & \text{if } 0\le x\le 1 \\ 2 & \text{if } 1<x\le 2 \\ x & \text{if } 2<x\le 5 \end{cases}$

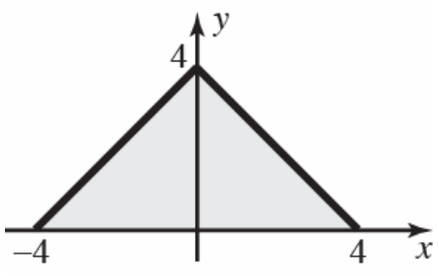
Solution:



$$\int_0^5 f(x)\,dx=\int_0^1 f(x)\,dx+\int_1^2 f(x)\,dx+\int_2^5 f(x)\,dx$$
$$=\frac{1}{2}(1)(2)+(1)(2)+\frac{1}{2}(2+5)(3)=\frac{27}{2}.$$

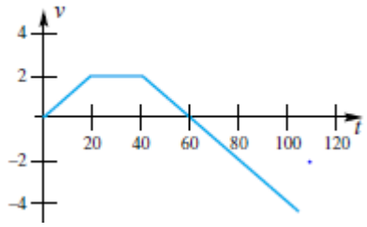
(b). $f(x)=4-|x|,\quad -4\le x\le 4.$

Solution:



$$\int_{-4}^4 f(x)\,dx=\int_{-4}^0 f(x)\,dx+\int_0^4 f(x)\,dx$$
$$=2\cdot\frac{1}{2}\cdot 4\cdot 4=16.$$

2. In the following graph, an object's velocity is graphed. Use this graph to determine the object's position at times $t=20,40,60,80,100$, assuming the object is at the origin at time $t=0$.



Sol:

t	20	40	60	80	100
$s(t)$	20	60	80	60	0

3. Let $f(x)$ be an odd function and $g(x)$ be an even function, and suppose that $\int_0^1 |f(x)|\,dx=\int_0^1 g(x)\,dx=3.$ Use geometric reasoning to calculate each of the following:

(a) $\int_{-1}^1 f(x)\,dx=0$ because this is an odd function.

(b) $\int_{-1}^1 |f(x)|\,dx=3+3=6$.

(c) $\int_{-1}^1 f^3(x)g(x)\,dx=0$ because $f^3(x)g(x)$ is an odd function.

(d) $\int_{-1}^1 xg(x)\,dx=0$ because $xg(x)$ is an odd function.

Exercises 5.3 The First Fundamental Theorem of Calculus

1. Suppose that $\int_0^1 f(x)\,dx=2, \int_1^2 f(x)\,dx=3, \int_0^1 g(x)\,dx=-1$ and $\int_0^2 g(x)\,dx=4$. Calculate each of the integrals in the following problems.

(a) $\int_0^1 [2f(s)+g(s)]\,ds$

Solution:

$$\int_0^1 [2f(s)+g(s)]\,ds=2\int_0^1 f(s)\,ds+\int_0^1 g(s)\,ds=2\cdot 2-1=3$$

(b) $\int_2^1 [2f(s)+5g(s)]\,ds$

Solution:

$$\int_2^1 [2f(s)+5g(s)]\,ds=2\int_2^1 f(s)\,ds+5\int_2^1 g(s)\,ds$$
$$=2\cdot(-3)+5\left[\int_0^1 g(s)\,ds-\int_0^2 g(s)\,ds\right]$$
$$=-6+5\cdot(-1-4)=-31$$

(c) $\int_1^1 [3f(s)+2g(s)]\,ds$

Solution: $\int_1^1 [3f(s)+2g(s)]\,ds=0$

2. Find $G'(x)$ for the following functions.

(a) $G(x)=\int_1^x (2t^2+\sqrt{t})\,dt$

Solution: $G'(x)=2x^2+\sqrt{x}$

(b) $G(x)=\int_x^{\frac{\pi}{4}} (s-2)\cot 2s\,ds,\ 0<x<\frac{\pi}{2}.$

Solution: $G'(x)=(2-x)\cot 2x$

(c) $G(x)=\int_1^{x^2}\sin t^2\,dt$.

Solution: $G'(x)=2x\sin x^4$

3. In the following problems, find the intervals on which the graph of $y=f(x),\ x\ge 0$ is (a) increasing, and (b) concave up.

(1) $f(x)=\int_1^x \frac{1+t}{1+t^2}\,dt$

Solution: $f'(x)=\frac{1+x}{1+x^2}$,

$$f''(x)=\frac{(1+x^2)-(1+x)2x}{\left(1+x^2\right)^2}=-\frac{x^2+2x-1}{\left(1+x^2\right)^2}.$$

So, $f(x)$ is increasing on $[0,\infty)$ and concave up on $(0,-1+\sqrt{2})$.

(2) $f(x)=\int_0^x (t+\sin t)\,dt$

Solution: $f'(x)=x+\sin x$, $f''(x)=1+\cos x$.
So, $f(x)$ is increasing on $(0,\infty)$ and concave up on $(0,\infty)$.

4. Find $\lim_{x\to 0}\frac{1}{x}\int_0^x \frac{1+t}{2+t}\,dt$.

Solution: By L'Hôspital rule, we have

$$\lim_{x\to 0}\frac{1}{x}\int_0^x \frac{1+t}{2+t}\,dt=\lim_{x\to 0}\frac{\frac{1+x}{2+x}}{1}=\frac{1}{2}.$$

Another Solution: Let $F(x)=\int_0^x \frac{1+t}{2+t}\,dt$. Then

$$\lim_{x\to 0}\frac{1}{x}\int_0^x \frac{1+t}{2+t}\,dt=\lim_{x\to 0}\frac{F(x)-F(0)}{x-0}$$
$$=F'(0)=\frac{1+0}{2+0}=\frac{1}{2}.$$

5. Find $\lim_{x\to 0}\frac{\int_0^x \sqrt{1+\sin t}\,dt}{x}$.

Solution: By L'Hôspital rule, we have

$$\lim_{x\to 0}\frac{\int_0^x \sqrt{1+\sin t}\,dt}{x}=\lim_{x\to 0}\sqrt{1+\sin x}=1$$

6. Find $\lim_{x\to 0^+}\frac{\int_0^x \sqrt{t}\cos t\,dt}{x^2}$.

Solution: By L'Hôspital rule, we have

$$\lim_{x\to 0^+}\frac{\int_0^x \sqrt{t}\cos t\,dt}{x^2}=\lim_{x\to 0^+}\frac{\sqrt{x}\cos x}{2x}=\infty.$$

7. Find $f(x)$ if $\int_1^x f(t)\,dt=2x-2$.

Solution: Differentiate both sides with respect to x.

$$\frac{d}{dx}\int_1^x f(t)\,dt=\frac{d}{dx}(2x-2)$$

gives $f(x)=2$.

8. Find $f(x)$ if $\int_0^{x^2} f(t)\,dt=\frac{1}{3}x^3,\ x>0$.

Solution: Differentiate both sides with respect to x.

$$\frac{d}{dx}\int_0^{x^2} f(t)\,dt=\frac{d}{dx}\left(\frac{1}{3}x^3\right)$$

gives $f(x^2)(2x)=x^2$.

Then $f(x^2)=\frac{x}{2}$, so $f(x)=\frac{\sqrt{x}}{2}$.

Exercises 5.4 The Second Fundamental Theorem of Calculus and the Method of Substitution.

1. In the following problems, use the Second Fundamental Theorem of Calculus to evaluate each definite integral.

(1) $\int_{-1}^{2}(3x^2-2x+3)\,dx$

Solution:

$$\int_{-1}^{2}(3x^2-2x+3)\,dx=(x^3-x^2+3x)\Big|_{-1}^{2}$$
$$=(8-4+6)-(-1-1-3)=15$$

(2) $\int_{1}^{4}\frac{1}{w^2}\,dw$

Solution:

$$\int_{1}^{4}\frac{1}{w^2}\,dw=\left(-\frac{1}{w}\right)\Big|_{1}^{4}=\left(-\frac{1}{4}\right)-(-1)=\frac{3}{4}$$

(3) $\int_{0}^{\frac{\pi}{2}}\cos x\,dx$

Solution:

$$\int_{0}^{\frac{\pi}{2}}\cos x\,dx=\sin x\Big|_{0}^{\pi/2}=1-0=1$$

2. Use the method of substitution to find each of the following definite integrals.

(1) $\int_{1}^{3}\frac{x^2+1}{\sqrt{x^3+3x}}\,dx$

Solution: Let $u=x^3+3x,\ du=(3x^2+3)dx$.

$$\int_{1}^{3}\frac{x^2+1}{\sqrt{x^3+3x}}\,dx=\frac{1}{3}\int_{4}^{36}\frac{1}{\sqrt{u}}\,du$$
$$=\frac{2}{3}u^{1/2}\Big|_{4}^{36}=\left(\frac{2}{3}\cdot 6\right)-\left(\frac{2}{3}\cdot 2\right)=\frac{8}{3}.$$

(2) $\int_{0}^{1}xe^{x^2}\,dx$

Solution: Let $u=x^2,\ du=2xdx$.

$$\int_{0}^{1}xe^{x^2}\,dx=\frac{1}{2}\int_{0}^{1}2xe^{x^2}\,dx=\frac{1}{2}\int_{0}^{1}e^u\,du$$
$$=\frac{1}{2}e^u\Big|_{0}^{1}=\frac{1}{2}(e-1).$$

(3) $\int_{0}^{\frac{\pi}{6}}\sin^3\theta\cos\theta\,d\theta$

Solution: Let $u=\sin\theta,\ du=\cos\theta d\theta$.

$$\int_{0}^{\frac{\pi}{6}}\sin^3\theta\cos\theta\,d\theta=\int_{0}^{\frac{1}{2}}u^3\,du=\frac{u^4}{4}\Big|_{0}^{1/2}=\frac{1}{64}.$$

(4) $\int_{0}^{1}x\cos^3(x^2)\sin(x^2)\,dx$

Solution: Let $u=\cos(x^2),\ du=-2x\sin(x^2)dx$.

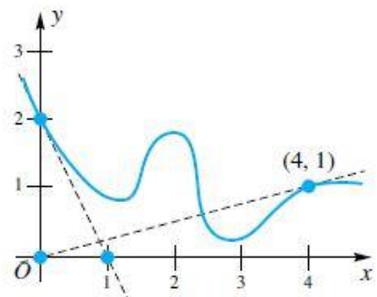
$$\int_{0}^{1}x\cos^3(x^2)\sin(x^2)\,dx$$
$$=-\frac{1}{2}\int_{1}^{\cos 1}u^3\,du=-\frac{1}{2}\left(\frac{u^4}{4}\right)\Big|_{1}^{\cos 1}$$
$$=\frac{1-\cos^4 1}{8}.$$

(5) $\int_{1}^{3}\frac{\ln x}{x}\,dx$

Solution: Let $u=\ln x,\ du=\frac{1}{x}dx$.

$$\int_{1}^{3}\frac{\ln x}{x}\,dx=\int_{0}^{\ln 3}u\,du=\frac{u^2}{2}\Big|_{0}^{\ln 3}=\frac{(\ln 3)^2}{2}$$

3 The following figure shows the graph of a function $f(x)$ that has a continuous third derivative. The dashed lines are tangent to the graph of $y=f(x)$ at the points $(0,2)$ and $(4,1)$. Based on what is shown, tell , if possible whether following integrals are positive, negative or zero.



(a) $\int_{0}^{4}f(x)\,dx$

Solution: For all $x\in[0,4]$, $f(x)>0$.

Thus $\int_{0}^{4}f(x)\,dx>0$

(b) $\int_{0}^{4}f'(x)\,dx$

Solution: $\int_{0}^{4}f'(x)\,dx=f(4)-f(0)=1-2=-1<0$

(c) $\int_{0}^{4}f''(x)\,dx$

Solution: $\int_{0}^{4}f''(x)\,dx=f'(4)-f'(0)=\frac{1}{4}-(-2)=\frac{9}{4}>0.$

(d) $\int_{0}^{4}f'''(x)\,dx$

Solution: Since f is concave up at 0, concave down at 4, $f''(4)<0$ and $f''(0)>0$. Therefore,

$$\int_{0}^{4}f''(x)\,dx=f''(4)-f''(0)<0.$$

Exercises 5.5 The Mean Value Theorem for Integrals and the Use of Symmetry.

1. In the following problems, use symmetry to help you evaluate the given integral.

(1) $\int_{-1}^{1} \frac{x^3}{(1+x^2)^4}\, dx$

Solution: Since the integrand is an odd function,

$\int_{-1}^{1} \frac{x^3}{(1+x^2)^4}\, dx = 0.$

(2) $\int_{-\pi/2}^{\pi/2} \frac{\sin x}{1+\cos x}\, dx$

Solution: Since the integrand is an odd function,

$\int_{-\pi/2}^{\pi/2} \frac{\sin x}{1+\cos x}\, dx = 0.$

(3) $\int_{-1}^{1} (1+x+x^2+x^3)\, dx$

Solution: $\int_{-1}^{1} (1+x+x^2+x^3)\, dx$

$= \int_{-1}^{1} (x+x^3)\, dx + \int_{-1}^{1} (1+x^2)\, dx$

$= 0 + 2\int_{0}^{1} (1+x^2)\, dx$

$= 2\left(x+\frac{x^3}{3}\right)\Big|_0^1 = \frac{8}{3}.$

(4) $\int_{-1}^{1} x e^{-4x^2}\, dx$

Solution: Since the integrand is an odd function,

$\int_{-1}^{1} x e^{-4x^2}\, dx = 0.$

(5) $\int_{-\pi/4}^{\pi/4} (|x|\sin^5 x + x^2 \tan x)\, dx$

Solution: Since the integrand is an odd function,

$\int_{-\pi/4}^{\pi/4} (|x|\sin^5 x + x^2 \tan x)\, dx = 0.$

(6) $\int_{-\pi}^{\pi} (\sin x + \cos x)^2\, dx$

Solution: $\int_{-\pi}^{\pi} (\sin x + \cos x)^2\, dx$

$= \int_{-\pi}^{\pi} (\sin^2 x + \cos^2 x + 2\sin x \cos x)\, dx$

$= \int_{-\pi}^{\pi} 1\, dx + \int_{-\pi}^{\pi} 2\sin x \cos x\, dx$

$= 2\pi + 0 = 2\pi.$

2. Calculate $\int_0^{4\pi} |\sin 2x|\, dx$

Solution: $\int_0^{4\pi} |\sin 2x|\, dx$

$= 4\int_0^{\pi} |\sin 2x|\, dx = 4\int_{-\pi/2}^{\pi/2} |\sin 2x|\, dx$

$= 8\int_0^{\pi/2} \sin 2x\, dx = 8\left(-\frac{1}{2}\cos 2x\right)\Big|_0^{\pi/2}$

$= -4(-1-1) = 8.$

3. Calculate $\int_0^{\pi/2} \sqrt{1-\sin 2x}\, dx$

Solution: $\int_0^{\pi/2} \sqrt{1-\sin 2x}\, dx$

$= \int_0^{\pi/2} |\sin x - \cos x|\, dx$

$= \int_0^{\pi/4} (\cos x - \sin x)\, dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\, dx$

$= (\sin x + \cos x)\Big|_0^{\pi/4} + (-\cos x - \sin x)\Big|_{\pi/4}^{\pi/2}$

$= 2\sqrt{2} - 2$

4. Let $f(x)$ be a continuous and increasing function on $[0,\infty)$ and let $F(x) = \frac{\int_0^x f(t)dt}{x}$, $x > 0$. Prove that $F(x)$ is increasing on $(0,\infty)$.

Proof: $F'(x) = \frac{xf(x) - \int_0^x f(t)dt}{x^2}$, $x > 0$.

By Mean Value Theorem for Integrals, for every $x > 0$, $\int_0^x f(t)dt = xf(\xi)$ for some $\xi \in (0,x)$.

Since f is increasing on $[0,\infty)$, $f(x) > f(\xi)$.

Then $F'(x) = \frac{xf(x) - xf(\xi)}{x^2} = \frac{f(x) - f(\xi)}{x} > 0,\quad x > 0$.

So $F(x)$ is increasing on $(0,\infty)$.

Exercises 7-1 Basic Integration Rules

1. In the following problems, perform the indicated integrations.

(1) $\int_0^1 x\sqrt{1-x^2}\,\mathrm{d}x$

Solution: Let $u=1-x^2$. So, $du=-2xdx$.
When $x=0, u=1$ *and when* $x=1, u=0$

$$\int_0^1 x\sqrt{1-x^2}dx=-\frac{1}{2}\int_1^0 u^{\frac{1}{2}}du=-\frac{1}{2}\cdot\frac{2}{3}u^{\frac{3}{2}}\Big|_1^0=\frac{1}{3}$$

Another Solution:

$$\int_0^1 x\sqrt{1-x^2}dx=-\frac{1}{2}\int_0^1\left(1-x^2\right)^{\frac{1}{2}}d\left(1-x^2\right)=-\frac{1}{2}\cdot\frac{2}{3}\left(1-x^2\right)^{\frac{3}{2}}\Big|_0^1=\frac{1}{3}$$

(2) $\int_0^{\pi/4}\frac{\tan z}{\cos^2 z}\,\mathrm{d}z$

Solution:

$$\int_0^{\pi/4}\frac{\tan z}{\cos^2 z}dz=\int_0^{\pi/4}\tan z\sec^2 z\,dz=\int_0^{\pi/4}\tan z\,d\tan z$$

$$=\frac{1}{2}\tan^2 z\Big|_0^{\pi/4}=\frac{1}{2}$$

(3) $\int_0^{\pi/4}\frac{\cos x}{1+\sin^2 x}\,\mathrm{d}x$

Solution:

$$\int_0^{\pi/4}\frac{\cos x}{1+\sin^2 x}dx=\int_0^{\pi/4}\frac{d\sin x}{1+\sin^2 x}=\arctan(\sin x)\Big|_0^{\pi/4}=\arctan\frac{\sqrt{2}}{2}$$

(4) $\int\frac{x^3}{x^4+4}\,\mathrm{d}x$

Solution:

$$\int\frac{x^3}{x^4+4}\,\mathrm{d}x=\frac{1}{4}\int\left(x^4+4\right)^{-1}d\left(x^4+4\right)=\frac{1}{4}\ln\left|x^4+4\right|+c$$

$$=\frac{1}{4}\ln(x^4+4)+c$$

(5) $\int\frac{\sin x-\cos x}{\sin x}\,\mathrm{d}x$

Solution:

$$\int\frac{\sin x-\cos x}{\sin x}\,\mathrm{d}x=\int(1-\frac{\cos x}{\sin x})dx=x-\int\frac{1}{\sin x}d\sin x$$

$$=x-\ln\left|\sin x\right|+c$$

(6) $\int\frac{1}{x^2+2x+5}\,\mathrm{d}x$

Solution:

$$\int\frac{1}{x^2+2x+5}\,\mathrm{d}x=\int\frac{1}{x^2+2x+1+4}\,\mathrm{d}x$$

$$=\int\frac{1}{(x+1)^2+2^2}\,\mathrm{d}(x+1)$$

$$=\frac{1}{2}\arctan(\frac{x+1}{2})+C$$

Exercises 7-2 Integration by Parts

1. In the following problems, use integration by parts to evaluate each integral.

(1) $\int xe^{3x}\mathrm{d}x$

Solution: Let $u=x$ and $v=\frac{1}{3}e^{3x}$. Then $du=dx$ and $dv=e^{3x}dx$.

$$\int xe^{3x}\mathrm{d}x=\int x\mathrm{d}\frac{1}{3}e^{3x}=\frac{1}{3}xe^{3x}-\frac{1}{3}\int e^{3x}dx=\frac{1}{3}xe^{3x}-\frac{1}{9}e^{3x}+C.$$

(2) $\int x\cos x\,\mathrm{d}x$

Solution: Let $u=x$ and $v=\sin x$. Then $du=dx$ and $dv=\cos xdx$.

$$\int x\cos x\mathrm{d}x=\int x\mathrm{d}\sin x=x\sin x-\int\sin xdx=x\sin x+\cos x+C$$

(3) $\int\frac{\ln x}{x^2}\,\mathrm{d}x$

Solution: Let $u=\ln x$ and $v=-\frac{1}{x}$. Then $du=\frac{1}{x}dx$ and $dv=\frac{1}{x^2}dx$.

$$\int\frac{\ln x}{x^2}\,\mathrm{d}x=\int\ln xd\left(-\frac{1}{x}\right)=-\frac{\ln x}{x}-\int-\frac{1}{x}(\frac{1}{x})dx$$

$$=-\frac{\ln x}{x}-\frac{1}{x}+C$$

(4) $\int t\arctan t\,\mathrm{d}t$

Solution: Let $u = \arctan t$ and $v = \frac{1}{2}t^2$. Then

$du = \frac{1}{1+t^2}dt$ and $dv = tdt$.

$$\int t\arctan t\ \mathrm{d}t = \int \arctan t\ \mathrm{d}\frac{1}{2}t^2$$
$$= \frac{1}{2}t^2\arctan t - \int \frac{1}{2}t^2 \cdot \frac{1}{1+t^2}dt$$
$$= \frac{1}{2}t^2\arctan t - \frac{1}{2}\int \frac{1+t^2-1}{1+t^2}dt$$
$$= \frac{1}{2}t^2\arctan t - \frac{1}{2}\int dt + \frac{1}{2}\int \frac{1}{1+t^2}dt$$
$$= \frac{1}{2}t^2\arctan t - \frac{1}{2}t + \frac{1}{2}\arctan t + C$$

(5) $\int_0^1 \ln(1+x^2)\,\mathrm{d}x$

Solution: Let $u = \ln(1+x^2)$ and $v = x$. Then

$du = \frac{2x}{1+x^2}dx$ and $dv = dx$.

$$\int_0^1 \ln(1+x^2)\,\mathrm{d}x = \left[x\ln(1+x^2)\right]\Big|_0^1 - \int_0^1 \frac{2x^2}{1+x^2}dx$$
$$= \ln 2 - \int_0^1 \frac{2x^2+2-2}{1+x^2}dx$$
$$= \ln 2 - \left(2x - 2\arctan x\right)\Big|_0^1$$
$$= \ln 2 - 2 + \frac{\pi}{2}.$$

(6) $\int_0^{1/2} \arcsin x\,\mathrm{d}x$

Solution: Let $u = \arcsin x$ and $v = x$. Then

$du = \frac{1}{\sqrt{1-x^2}}dx$ and $dv = dx$.

$$\int_0^{1/2} \arcsin x\,\mathrm{d}x = \left(x\arcsin x\right)\Big|_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}dx$$
$$= \frac{1}{2}\cdot\frac{\pi}{6} - \left(-\sqrt{1-x^2}\right)\Big|_0^{1/2} = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$$

2. In the following problems, apply integration by parts twice to evaluate each integral.

(1) $\int x^2 e^x \mathrm{d}x$

Solution:

$$\int x^2 e^x \mathrm{d}x = \int x^2 \mathrm{d}e^x = x^2e^x - \int e^x dx^2 = x^2e^x - \int 2xe^x dx$$
$$= x^2e^x - 2\int xde^x = x^2e^x - 2(xe^x - \int e^x dx)$$
$$= x^2e^x - 2xe^x + 2e^x + C$$

(2) $\int e^{at}\sin t\,\mathrm{d}t$

Solution:

$$\int e^{at}\sin tdt = \frac{1}{a}\int \sin tde^{at} = \frac{1}{a}e^{at}\sin t - \frac{1}{a}\int e^{at}\cos tdt$$
$$= \frac{1}{a}e^{at}\sin t - \frac{1}{a}\left[\frac{1}{a}e^{at}\cos t + \frac{1}{a}\int e^{at}\sin tdt\right]$$
$$= \frac{1}{a}e^{at}\sin t - \frac{1}{a^2}e^{at}\cos t - \frac{1}{a^2}\int e^{at}\sin tdt$$

Therefore, $\left(1+\frac{1}{a^2}\right)\int e^{at}\sin tdt = \frac{1}{a}e^{at}\sin t - \frac{1}{a^2}e^{at}\cos t$.

That is, $\int e^{at}\sin tdt = \frac{a\sin t - \cos t}{a^2+1}e^{at} + C$.

3. Assume that $f''(x)$ is continuous on [0, 1] and $f(0)=1, f(1)=3, f'(1)=7$. Find $\int_0^1 xf''(x)\mathrm{d}x$.

Solution:

$$\int_0^1 xf''(x)\mathrm{d}x = \int_0^1 x\mathrm{d}f'(x) = [xf'(x)]_0^1 - \int_0^1 f'(x)\mathrm{d}x$$
$$= f'(1) - f(1) + f(0)$$
$$= 7 - 3 + 1 = 5$$

Exercises 7-3 Some Trigonometric Integrals

1. In the following problems, perform the indicated integrations.

(1) $\int \sin^3 x\,\mathrm{d}x$

Solution:

$$\int \sin^3 x\,\mathrm{d}x = \int \sin x(1-\cos^2 x)dx$$
$$= \int \sin xdx - \int \sin x\cos^2 xdx$$
$$= -\cos x + \frac{1}{3}\cos^3 x + C$$

(2) $\int_0^{\pi/2} \cos^5\theta\,\mathrm{d}\theta$

Solution:

$$\int_0^{\pi/2} \cos^5\theta\,\mathrm{d}\theta = \int_0^{\pi/2} (1-\sin^2\theta)^2\cos\theta\,\mathrm{d}\theta$$
$$= \int_0^{\pi/2} (1-2\sin^2\theta+\sin^4\theta)\cos\theta d\theta$$
$$= \left[\sin\theta - \frac{2}{3}\sin^3\theta + \frac{1}{5}\sin^5\theta\right]_0^{\pi/2}$$
$$= \frac{8}{15}$$

(3) $\int_0^{\pi/2} \sin^4\theta\,\mathrm{d}\theta$

Solution:

$$\int_0^{\pi/2} \sin^4\theta\,\mathrm{d}\theta = \frac{3}{4}\int_0^{\pi/2} \sin^2\theta\,\mathrm{d}\theta$$
$$= \frac{3}{4}\cdot\frac{1}{2}\int_0^{\pi/2} \sin^0\theta\,\mathrm{d}\theta = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16}.$$

(4) $\int \sin 4y \cos 5y \, \mathrm{d}y$

Solution:

$$\int \sin 4y \cos 5y \, \mathrm{d}y = \frac{1}{2}\int [\sin 9y + \sin(-y)]dy$$

$$= \frac{1}{2}\int (\sin 9y - \sin y)dy = \frac{1}{2}(-\frac{1}{9}\cos 9y + \cos y) + C$$

$$= \frac{1}{2}\cos y - \frac{1}{18}\cos 9y + C$$

(5) $\int \tan^3 x \, \mathrm{d}x$

Solution:

$$\int \tan^3 x \, dx = \int \tan x \left(\tan^2 x\right) dx = \int \tan x \left(\sec^2 x - 1\right) dx$$

$$= \int \tan x \, d \tan x - \int \tan x \, dx = \frac{1}{2}\tan^2 x + \ln|\cos x| + C$$

(6) $\int \tan^3 x \sec x \mathrm{d}x$

Solution:

$$\int \tan^3 x \sec x dx = \int \tan^2 x \left(\tan x \sec x\right) dx = \int \left(\sec^2 x - 1\right) d \sec x$$

$$= \int \sec^2 x d \sec x - \int 1 d \sec x = \frac{1}{3}\sec^3 x - \sec x + C$$

(7) $\int \sin^{-4} x \cos^2 x \mathrm{d}x$

Solution:

$$\int \sin^{-4} x \cos^2 x \mathrm{d}x = \int \sin^{-4} x \cos^4 x \sec^2 x \mathrm{d}x = \int \tan^{-4} x \, \mathrm{d}\tan x$$

$$= -\frac{1}{3}\tan^{-3} x + C$$

Exercises 7-4 Rationalizing Substitutions

1. In the following problems, perform the indicated integrations.

(1) $\int \frac{t \, \mathrm{d}t}{\sqrt{3t+4}}$

Solution: Let $u = \sqrt{3t+4}$. Then $t = \frac{u^2 - 4}{3}$ and $dt = \frac{2}{3}udu$.

$$\int \frac{t \, \mathrm{d}t}{\sqrt{3t+4}} = \int \frac{\frac{1}{3}(u^2-4)\frac{2}{3}udu}{u} = \frac{2}{9}\int (u^2 - 4)du$$

$$= \frac{2}{27}u^3 - \frac{8}{9}u + C$$

$$= \frac{2}{27}(3t+4)^{3/2} - \frac{8}{9}(3t+4)^{1/2} + C$$

(2) $\int_1^2 \frac{\mathrm{d}t}{\sqrt{t}+e}$

Solution: Let $u = \sqrt{t}$. Then $t = u^2$ and $dt = 2udu$. When t=1, $u = 1$. When $t = 2$, $u = \sqrt{2}$.

$$\int_1^2 \frac{\mathrm{d}t}{\sqrt{t}+e} = \int_1^{\sqrt{2}} \frac{2udu}{u+e} = 2\int_1^{\sqrt{2}} \frac{u+e-e}{u+e}du$$

$$= 2\int_1^{\sqrt{2}} du - 2\int_1^{\sqrt{2}} \frac{e}{u+e}du$$

$$= 2u\Big|_1^{\sqrt{2}} - 2e\ln|u+e|\Big|_1^{\sqrt{2}}$$

$$= 2\sqrt{2} - 2 - 2e\ln\left(\frac{\sqrt{2}+e}{1+e}\right)$$

(3) $\int \frac{x^2 \mathrm{d}x}{\sqrt{16-x^2}}$

Solution: Let $x = 4\sin t, t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $dx = 4\cos t dt$ and $\cos x \geq 0$.

$$\int \frac{x^2 \mathrm{d}x}{\sqrt{16-x^2}} = \int \frac{16\sin^2 t}{4|\cos t|}(4\cos t)dt$$

$$= 16\int \sin^2 t dt = 8\int (1 - \cos 2t)dt$$

$$= 8t - 4\sin 2t + C = 8t - 8\sin t \cos t + C$$

$$= 8\arcsin(\frac{x}{4}) - \frac{x\sqrt{16-x^2}}{2} + C$$

(4) $\int \frac{\mathrm{d}x}{(x^2+4)^{3/2}}$

Solution: Let $x = 2\tan t, t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $dx = 2\sec^2 t dt$ and $\sec x > 0$.

$$\int \frac{\mathrm{d}x}{(x^2+4)^{3/2}} = \int \frac{2\sec^2 t dt}{(4\sec^2 t)^{3/2}} = \frac{1}{4}\int \cos t dt$$

$$= \frac{1}{4}\sin t + C = \frac{x}{4\sqrt{x^2+4}} + C$$

(5) $\int_2^3 \frac{\sqrt{t^2-1}}{t^3} \mathrm{d}t$

Solution: Let $t = \sec x$. So, $dt = \sec x \tan x dx$. When $t = \sec x = 3$, $\cos x = \frac{1}{3}$. So, $x = \arccos\frac{1}{3} < \frac{\pi}{2}$. When $t = 2$, $x = \arccos\frac{1}{2} = \frac{\pi}{3} < \arccos\frac{1}{3}$. Therefore, $\frac{\pi}{3} \leq x \leq \arccos\frac{1}{3} < \frac{\pi}{2}$ and $\sqrt{t^2-1} = \tan x$.

$$\int_2^3 \frac{\sqrt{t^2-1}}{t^3} \mathrm{d}t = \int_{\pi/3}^{\arccos\frac{1}{3}} \frac{\tan x}{\sec^3 x}\sec x \tan x dx$$

$$= \int_{\pi/3}^{\arccos\frac{1}{3}} \sin^2 x dx = \int_{\pi/3}^{\arccos\frac{1}{3}} (\frac{1}{2} - \frac{1}{2}\cos 2x)dx$$

$$= \left(\frac{1}{2}x - \frac{1}{4}\sin 2x\right)\Bigg|_{\pi/3}^{\arccos\frac{1}{3}} = \left(\frac{1}{2}x - \frac{1}{2}\sin x \cos x\right)\Bigg|_{\pi/3}^{\arccos\frac{1}{3}}$$

Since $\cos\left(\arccos\frac{1}{3}\right) = \frac{1}{3}$ and $\sin\left(\arccos\frac{1}{3}\right) = \frac{2\sqrt{2}}{3}$,

$$\int_2^3 \frac{\sqrt{t^2-1}}{t^3} \mathrm{d}t = \left(\frac{1}{2}x - \frac{1}{2}\sin x \cos x\right)\Bigg|_{\pi/3}^{\arccos\frac{1}{3}}$$

$$= \frac{1}{2}\arccos\frac{1}{3} - \frac{1}{2}\cdot\frac{2\sqrt{2}}{3}\cdot\frac{1}{3} - \frac{\pi}{6} + \frac{1}{2}\cdot\frac{\sqrt{3}}{2}\cdot\frac{1}{2}$$

$$= \frac{1}{2}\arccos\frac{1}{3} - \frac{\sqrt{2}}{9} - \frac{\pi}{6} + \frac{\sqrt{3}}{8}.$$

2. In the following problems, use the method of completing the square, along with a trigonometric substitution if needed, to evaluate each integral.

(1) $\int \frac{2x-1}{\sqrt{x^2+4x+5}}\,\mathrm{d}x$

Solution:

$$\int \frac{2x-1}{\sqrt{x^2+4x+5}}\,dx = \int \frac{2(x+2)-5}{\sqrt{(x+2)^2+1}}d(x+2)$$

$$= \int \frac{2(x+2)}{\sqrt{(x+2)^2+1}}d(x+2) - 5\int \frac{1}{\sqrt{(x+2)^2+1}}d(x+2)$$

$$= 2\sqrt{(x+2)^2+1} - 5\ln\left|\sqrt{x^2+4x+5}+x+2\right| + C$$

(2) $\int \sqrt{5-4x-x^2}\,\mathrm{d}x$

Solution: $\int \sqrt{5-4x-x^2}\,\mathrm{d}x = \int \sqrt{9-(x+2)^2}\,d(x+2)$

Let $x+2=3\sin t$, $t\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. Then

$d(\mathrm{x}+2)=3\cos t dt$ and $\cos t \ge 0$.

$$\int \sqrt{9-(x+2)^2}\,d(x+2) = 9\int \cos^2 t dt = \frac{9}{2}\int (1+\cos 2t)dt$$

$$= \frac{9}{2}(t+\frac{1}{2}\sin 2t) + C = \frac{9}{2}(t+\sin t\cos t) + C$$

$$= \frac{9}{2}\arcsin(\frac{x+2}{3}) + \frac{x+2}{2}\sqrt{5-4x-x^2} + C$$

Exercises 7-5 Integration of Rational Functions Using Partial Fractions

1. In the following problems, use the method of partial fraction decomposition to perform the required integration.

(1) $\int \frac{3}{x^2-1}\,\mathrm{d}x$

Solution: Since

$$\frac{3}{x^2-1} = \frac{3}{(x+1)(x-1)} = \frac{-3}{2(x+1)} + \frac{3}{2(x-1)},$$

$$\int \frac{3}{x^2-1}\,\mathrm{d}x = -\frac{3}{2}\int \frac{1}{x+1}dx + \frac{3}{2}\int \frac{1}{x-1}dx$$

$$= -\frac{3}{2}\ln|x+1| + \frac{3}{2}\ln|x-1| + C$$

$$= \frac{3}{2}\ln\left|\frac{x-1}{x+1}\right| + C$$

(2) $\int \frac{3x-13}{x^2+3x-10}\,\mathrm{d}x$

Solution: Since

$$\frac{3x-13}{x^2+3x-10} = \frac{3x-13}{(x+5)(x-2)} = \frac{4}{x+5} - \frac{1}{x-2},$$

$$\int \frac{3x-13}{x^2+3x-10}\,\mathrm{d}x = 4\int \frac{1}{x+5}dx - \int \frac{1}{x-2}dx$$

$$= 4\ln|x+5| - \ln|x-2| + C$$

(3) $\int \frac{1}{x^2+2}\,\mathrm{d}x$

Solution: $\int \frac{1}{x^2+2}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\arctan\frac{x}{\sqrt{2}} + C$

(4) $\int \frac{x^3}{x^2+1}\,\mathrm{d}x$

Solution:

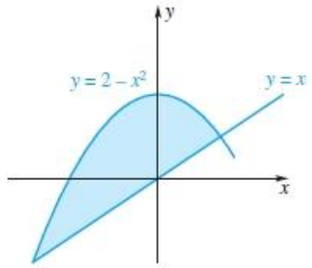
$$\int \frac{x^3}{x^2+1}\,\mathrm{d}x = \int \frac{x^3+x-x}{x^2+1}\,\mathrm{d}x = \int x - \frac{x}{x^2+1}\,\mathrm{d}x$$

$$= \frac{x^2}{2} - \frac{1}{2}\int \frac{1}{x^2+1}\,\mathrm{d}(x^2+1) = \frac{x^2}{2} - \ln\sqrt{x^2+1} + C.$$

Exercises 6-1 The Area of a Plane Region

1. In the following problems, use the three-step procedure (slice, approximation and integrate) to set up and evaluate an integral for the area of the indicated region.

(1)

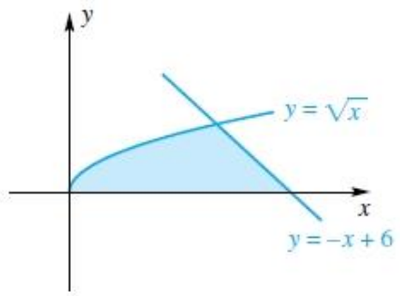


Solution: By solving the equation $2-x^2=x$, we can get $x=-2,1$. Slicing vertically, we can get

$$\Delta A \approx [(2-x^2)-x]\Delta x=(-x^2-x+2)\Delta x.$$

$$A=\int_{-2}^{1}(-x^2-x+2)dx=-\frac{1}{3}x^3-\frac{1}{2}x^2+2x\Big|_{-2}^{1}=\frac{9}{2}.$$

(2)



Solution: By solving the equation, $y^2=6-y$, we can get $y=-3,2$. Slice horizontally,

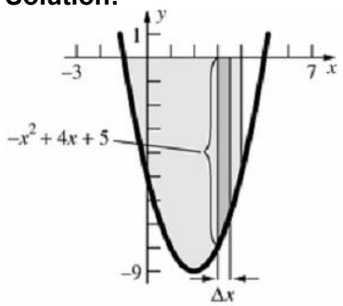
$$\Delta A \approx [(6-y)-y^2]\Delta y=(-y^2-y+6)\Delta y.$$

$$A=\int_0^2(-y^2-y+6)dy=-\frac{1}{3}y^3-\frac{1}{2}y^2+6y\Big|_0^2=\frac{22}{3}.$$

2. In the following problems, sketch the region bounded by the graphs of the given equations, show a typical slice, approximate its area, set up an integral, and calculate the area of the region.

(1) $y=x^2-4x-5$, $y=0$, between $x=-1$ and $x=4$.

Solution:

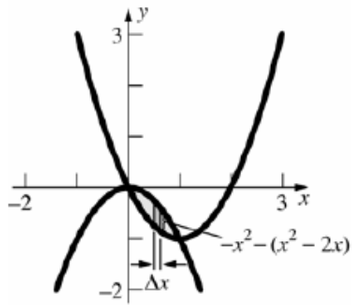


$$\Delta A \approx -(x^2-4x-5)\Delta x.$$

$$A=\int_{-1}^{4}(-x^2+4x+5)dx=[-\frac{1}{3}x^3+2x^2+5x]_{-1}^{4}=\frac{100}{3}.$$

(2) $y=x^2-2x$, $y=-x^2$.

Solution:



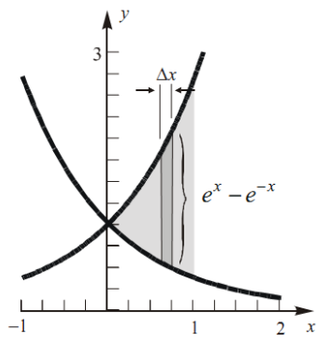
$$\Delta A \approx (2x-2x^2)\Delta x.$$

Solve the equation $-x^2=x^2-2x$ to get $x=0$ and $x=1$

$$A=\int_0^1(2x-2x^2)dx=[x^2-\frac{2}{3}x^3]_0^1=\frac{1}{3}.$$

(3) $y=e^x$, $y=e^{-x}$, between $x=0$ and $x=1$.

Solution:



$$\Delta A \approx (e^x-e^{-x})\Delta x.$$

$$A=\int_0^1(e^x-e^{-x})dx=(e^x+e^{-x})\Big|_0^1=e+\frac{1}{e}-2.$$

3. Consider the curve $y=\frac{1}{x^2}$ for $1\le x\le 6$.

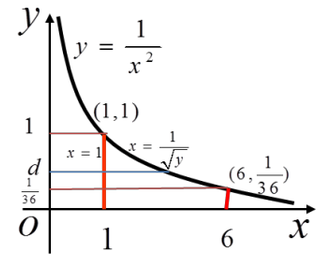
(a) Calculate the area under this curve.

(b) Determine *c* so that the line $x=c$ bisects the area of part (a).

(c) Determine *d* so that the line $y=d$ bisects the area of part (a).

Solution: a. $A=\int_1^6 x^{-2}dx=[\frac{-1}{x}]_1^6=\frac{5}{6}.$

b. Find c so that $\int_1^c x^{-2}dx=\frac{5}{12}.$



$$\int_1^c x^{-2}dx=[\frac{-1}{x}]_1^c=1-\frac{1}{c}.$$

$$1-\frac{1}{c}=\frac{5}{12}, c=\frac{12}{7}.$$

The line $x=\frac{12}{7}$ bisects the area.

c. Slicing the region horizontally, the area is

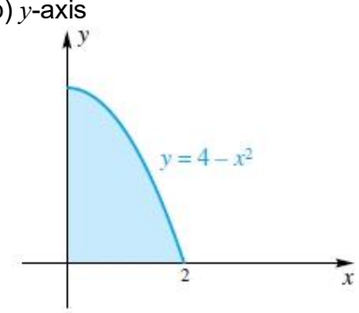
$$\int_{1/36}^{1}\left(\frac{1}{\sqrt{y}}-1\right)dy+\frac{5}{36}.$$ Since $\frac{5}{36}<\frac{5}{12}$,

$$\int_d^1\left(\frac{1}{\sqrt{y}}-1\right)dy=\frac{5}{12}, d-2\sqrt{d}+\frac{7}{12}=0, d=\frac{17}{12}-\frac{\sqrt{15}}{3}.$$

Exercises 6-2&6-3 Volumes of Solids: Slabs, Disks, Washers and Shells

1. Find the volume of the solid generated when the indicated region is revolved about the specified axis.

(a) x-axis (b) y-axis



Solution:

(a) $\Delta V \cong \pi(4-x^2)^2\Delta x.$

$$V=\int_0^2 \pi(4-x^2)^2\mathrm{d}x=\frac{256\pi}{15}.$$

(b) Since $y=4-x^2$, we have that $x=\sqrt{4-y}$.

$$\Delta V \cong \pi(\sqrt{4-y})^2\Delta y.$$

$$V=\int_0^4 \pi(4-y)\mathrm{d}y=8\pi.$$

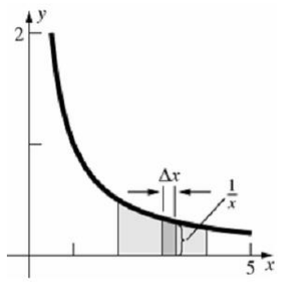
2. In the following problems, sketch the region *R* bounded by the graphs of the given equations, and show a typical vertical slice. Then find the volume of the solid generated by revolving *R* about the x-axis.

(1) $y=\frac{1}{x}$, $x=2$, $x=4$ and $y=0$

Solution:

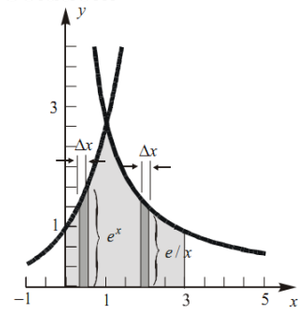
$$\Delta V \cong \pi\frac{1}{x^2}\Delta x.$$

$$V=\int_2^4 \pi\frac{1}{x^2}\mathrm{d}x=\pi[-\frac{1}{x}]_2^4=\frac{\pi}{4}.$$



(2) $y=e^x$, $y=\frac{e}{x}$, $y=0$ between $x=0$ and $x=3$

Solution:



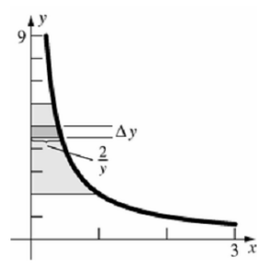
$$V=\int_0^1 \pi e^{2x}\mathrm{d}x+\int_1^3 \pi\frac{e^2}{x^2}\mathrm{d}x$$

$$=\pi[\frac{1}{2}e^{2x}]_0^1+\pi[-\frac{1}{x}e^2]_1^3=\frac{7\pi e^2}{6}-\frac{\pi}{2}.$$

3. In the following problems, sketch the region *R* bounded by the graphs of the given equations, and show a typical horizontal slice. Find the volume of the solid generated by revolving *R* about the y-axis.

(1) $x=\frac{2}{y}$, $y=2$, $y=6$ and $x=0$

Solution:

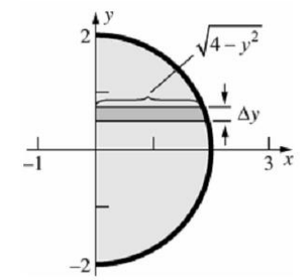


$$\Delta V \cong \pi(\frac{2}{y})^2\Delta y.$$

$$V=\int_2^6 \pi\frac{4}{y^2}\mathrm{d}y=4\pi[-\frac{1}{y}]_2^6=\frac{4\pi}{3}.$$

(2) $x=\sqrt{4-y^2}$, $x=0$

Solution:

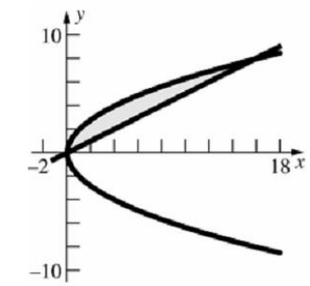


$$\Delta V \approx \pi(\sqrt{4-y^2})^2\Delta y.$$

$$V=\int_{-2}^{2} \pi(4-y^2)\mathrm{d}y$$

$$=\pi[4y-\frac{1}{3}y^3]_{-2}^2=\frac{32\pi}{3}.$$

4.



Find the volume of the solid generated by revolving about the x-axis the region bounded by the line $x-2y=0$ and the parabola $y^2=4x$.

Solution:

To find the intersections,

solve $\frac{x}{2}=2\sqrt{x}$. $\Rightarrow x=0$ or $x=16$.

Therefore,

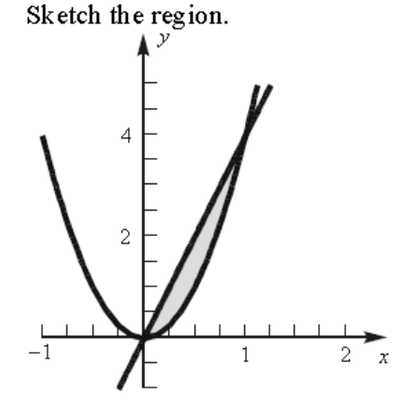
$$\Delta V \approx \pi(2\sqrt{x})^2\Delta x-\pi(\frac{x}{2})^2\Delta x.$$

$$V=\int_0^{16} \pi(4x-\frac{1}{4}x^2)\mathrm{d}x=\pi[2x^2-\frac{1}{12}x^3]_0^{16}=\frac{512\pi}{3}.$$

5. Use the method of shells to find the volume of the solid generated by revolving about the y-axis the region bounded by the line $y=4x$ and the parabola $y=4x^2$.

Solution:

Sketch the region.



To find the intersection points, solve $4x=4x^2$, x = 0,1.

$\Delta V \approx 2\pi x(4x-4x^2)\Delta x.$

$$V=\int_0^1 2\pi x(4x-4x^2)\mathrm{d}x = 2\pi[\frac{4}{3}x^3-x^4]_0^1=\frac{2\pi}{3}.$$

Exercises 6-4 Length of a Plane Curve

1. In the following problems, find the length of the indicated curve. (No. 1, 8)

(1) $y=4x^{3/2}$ between $x=\frac{1}{3}$ and $x=5$

Solution:

$f(x)=4x^{3/2}, f'(x)=6x^{1/2}.$

$$L=\int_{1/3}^5\sqrt{1+(6x^{1/2})^2}\mathrm{d}x=\int_{1/3}^5\sqrt{1+36x}\mathrm{d}x=$$

$$=\frac{1}{36}\cdot\frac{2}{3}(1+36x)^{3/2}\Big|_{1/3}^5=\frac{1}{54}(181\sqrt{181}-13\sqrt{13}).$$

(2) $x=e^t\cos t, y=e^t\sin t$ for $0\le t\le 1$.

Solution:

$\frac{dx}{dt}=e^t\cos t-e^t\sin t, \frac{dy}{dt}=e^t\sin t+e^t\cos t.$

$$L=\int_0^1\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}dt$$

$$=\int_0^1\sqrt{(e^t\cos t-e^t\sin t)^2+(e^t\sin t+e^t\cos t)^2}$$

$$=\int_0^1\sqrt{2(e^{2t}\cos^2 t+e^{2t}\sin^2 t)}dt=\sqrt{2}\int_0^1 e^t dt=\sqrt{2}(e-1).$$

2. Find the length of the curve $x=t-\sin t, y=1-\cos t,$ $0\le t\le 4\pi$.

Solution:

$f'(t)=1-\cos t, g'(t)=\sin t.$

$$L=\int_0^{4\pi}\sqrt{2-2\cos t}dt=2\int_0^{4\pi}\left|\sin(\frac{t}{2})\right|dt$$

$$=4\int_0^{2\pi}\sin(\frac{t}{2})dt=-8\cos\frac{t}{2}\Big|_0^{2\pi}=16.$$

Exercises 6-7 Probability and Random Variables

In Problems 1-2, a PDF for a continuous random variable X is given. Use the PDF to find $P(\mathrm{X}\ge 2)$ and $E(\mathrm{X})$.

1. $f(\mathrm{x})=\begin{cases}\frac{1}{40}, \text{if} -20\le x\le 20\\ 0, \text{otherwise}\end{cases}$

Solution:

$$P(\mathrm{X}\ge 2)=\int_2^{20}\frac{1}{40}dx=\frac{20-2}{40}=0.45.$$

$$E(\mathrm{X})=\int_{-20}^{20}\frac{x}{40}dx=\frac{x^2}{80}\Big|_{-20}^{20}=5-5=0.$$

2. $f(\mathrm{x})=\begin{cases}\frac{3}{64}\mathrm{x}^2(4-\mathrm{x}), \text{if } 0\le x\le 4\\ 0, \text{otherwise}\end{cases}$

Solution:

$$P(X\ge 2)=\int_2^4\frac{3}{64}x^2(4-x)dx=\frac{1}{64}\left(4x^3-\frac{3}{4}x^4\right)\Big|_2^4=0.6875$$

.

$$E(\mathrm{X})=\int_0^4\frac{3}{64}x^3(4-x)dx=\frac{3}{64}\left(x^4-\frac{1}{5}x^5\right)\Big|_0^4=2.4.$$

3. A continuous random variable X is said to have a **uniform distribution** on the interval $[a,b]$ if the PDF has the form

$$f(\mathrm{x})=\begin{cases}\frac{1}{b-a}, \text{if } a\le x\le b\\ 0, \text{otherwise}\end{cases}$$

(a) Find the probability that the value of X is closer to a than it is to b.

(b) Find the expected value of X.

(c) Find the CDF of X.

Solution: (a)

$$P\left(x<\frac{a+b}{2}\right)=P\left(x\le\frac{a+b}{2}\right)=\int_a^{\frac{a+b}{2}}\frac{1}{b-a}dx$$

$$=\frac{1}{b-a}\left(\frac{a+b}{2}-a\right)=0.5$$

(b)

$$E(\mathrm{X})=\int_a^b\frac{x}{b-a}dx=\frac{x^2}{2(b-a)}\Big|_a^b=\frac{b+a}{2}$$

(c)

$$F(x)=\int_a^x\frac{1}{b-a}dt=\frac{x-a}{b-a}$$

Exercises 7-7 First-Order Linear Differential Equations

1. Solve the following differential equations

(1) $(x+1)\dfrac{dy}{dx}+y=x^2-1$

Solution:

The left hand side is already an exact derivative.

$D[y(x+1)]=x^2-1$

$y=\dfrac{x^3-3x+C}{3(x+1)}$

(2) $y'+y\tan x=\sec x$

Solution:

Let $P(x)=\tan x$ and $Q(x)=\sec x$.

Then the integrating factor is

$I(x)=e^{\int P(x)dx}=e^{\int \tan x dx}=\sec x$.

Therefore,

$y=\dfrac{1}{I(x)}\int Q(x)I(x)dx=\dfrac{1}{\sec x}\int(\sec x\sec x)dx$

$=\cos x(\tan x+C)=\sin x+C\cos x$

(3) $\dfrac{dy}{dx}+\dfrac{y}{x}=\dfrac{1}{x}$

Solution:

Let $P(x)=\dfrac{1}{x}$ and $Q(x)=\dfrac{1}{x}$.

Then the integrating factor is $I(x)=e^{\int P(x)dx}=e^{\int \tan x dx}=x$.

Therefore,

$y=\dfrac{1}{I(x)}\int Q(x)I(x)dx=\dfrac{1}{x}\int 1dx=\dfrac{1}{x}(x+C)=1+\dfrac{C}{x}$

(4) $y'=e^{2x}-3y;\ y=1$ when $x=0$

Solution:

Let $P(x)=3$ and $Q(x)=e^{2x}$.

Then the integrating factor is $I(x)=e^{\int P(x)dx}=e^{\int 3dx}=e^{3x}$.

Therefore,

$y=\dfrac{1}{I(x)}\int Q(x)I(x)dx=\dfrac{1}{e^{3x}}\int e^{5x}\,dx=\dfrac{e^{-3x}}{5}(e^{5x}+C)$

$=\dfrac{e^{2x}+Ce^{-3x}}{5}$

Since $y(0)=\dfrac{e^0+Ce^0}{5}=\dfrac{1+C}{5}=1$, $C=4$. So,

$y=\dfrac{e^{2x}+4e^{-3x}}{5}$.

Exercises 8-1 Improper Integrals: Infinite Limits of Integration

1. In the following problems, evaluate each improper integral or show that it diverges.

(1) $\int_1^\infty 2xe^{-x^2}dx$

Solution:

$\int_1^\infty 2xe^{-x^2}dx=[-e^{-x^2}]_1^\infty=0-(-e^{-1})=\dfrac{1}{e}$

(2) $\int_2^\infty \dfrac{\ln x}{x^2}dx$

Solution:

let $u=\ln x, du=\dfrac{1}{x}dx, dv=\dfrac{1}{x^2}dx, v=-\dfrac{1}{x}$

$\int_2^\infty \dfrac{\ln x}{x^2}dx=\lim_{b\to\infty}\int_2^b \dfrac{\ln x}{x^2}dx$

$=\lim_{b\to\infty}[-\dfrac{\ln x}{x}]_2^b+\lim_{b\to\infty}\int_2^b \dfrac{1}{x^2}dx$

$=\lim_{b\to\infty}[-\dfrac{\ln x}{x}-\dfrac{1}{x}]_2^b=\dfrac{\ln 2+1}{2}$